Wolf Creek
Nuclear Operating Corporation

Dustin T. Hamman
Director Nuclear and Regulatory Affairs

March 5, 2026
001257

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Subject: Docket No. 50-482: Wolf Creek Generating Station Changes to Technical Specification Bases – Revisions 97 and 98

Commissioners and Staff:

The Wolf Creek Generating Station (WCGS) Unit 1 Technical Specifications (TS), Section 5.5.14, "Technical Specifications (TS) Bases Control Program," provides the means for making changes to the Bases without prior Nuclear Regulatory Commission (NRC) approval. In addition, TS Section 5.5.14 requires that changes made without NRC approval be provided to the NRC on a frequency consistent with 10 CFR 50.71(e). The Enclosure provides those changes made to the WCGS TS Bases (Revisions 97 and 98) under the provisions to TS Section 5.5.14 and a List of Effective Pages. This submittal reflects changes from January 1, 2025, through December 31, 2025.

This letter contains no commitments. If you have any questions concerning this matter, please contact me at (620) 364-4204.

Sincerely,

Dustin T. Hamman

DTH/jkt

Enclosure: Wolf Creek Generating Station Changes to the Technical Specification Bases

cc: A. N. Agrawal (NRC), w/e
S. S. Lee (NRC), w/e
J. D. Monninger (NRC), w/e
Senior Resident Inspector (NRC), w/e
WC Licensing Correspondence, w/e – RA 26-001257

Enclosure to 001257

Wolf Creek Generating Station
Changes to the Technical Specification Bases

(64 pages)

TECHNICAL SPECIFICATION BASES

Wolf Creek Generating Station, Unit 1

Summary of Revision 97:

1) Revised TS Bases pages B 3.3.1-56, B 3.3.1-57, B 3.3.2-50, and B 3.3.2-52, as part of incorporation of License Amendment 243. Amendment 243 revises technical specification definitions for engineered safety features (ESF) response time and reactor trip system (RTS) response time that are referenced in surveillance requirements (SRs). These changes are based on Technical Specifications Task Force (TSTF) traveler TSTF-569, Revision 2, “Revise Response Time Testing Definition,” dated June 25, 2019.

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – Title Page Technical Specification			
Cover Page			
Title Page			
TAB – Table of Contents			
i	81	DRR 19-1027	10/28/19
ii	81	DRR 19-1027	10/28/19
iii	81	DRR 19-1027	10/28/19
TAB – B 2.0 SAFETY LIMITS (SLs)			
B 2.1.1-1	0	Amend. No. 123	12/18/99
B 2.1.1-2	14	DRR 03-0102	2/12/03
B 2.1.1-3	14	DRR 03-0102	2/12/03
B 2.1.1-4	14	DRR 03-0102	2/12/03
B 2.1.2-1	84	DRR 20-0400	08/18/20
B 2.1.2-2	84	DRR 20-0400	08/18/20
B 2.1.2-3	81	DRR 19-1027	10/28/19
TAB – B 3.0 LIMITING CONDITION FOR OPERATION (LCO) APPLICABILTY			
B 3.0-1	81	DRR 19-1027	10/28/19
B 3.0-2	0	Amend. No. 123	12/18/99
B 3.0-3	81	DRR 19-1027	10/28/19
B 3.0-4	81	DRR 19-1027	10/28/19
B 3.0-5	81	DRR 19-1027	10/28/19
B 3.0-6	81	DRR 19-1027	10/28/19
B 3.0-7	81	DRR 19-1027	10/28/19
B 3.0-8	81	DRR 19-1027	10/28/19
B 3.0-9	81	DRR 19-1027	10/28/19
B 3.0-10	81	DRR 19-1027	10/28/19
B 3.0-11	81	DRR 19-1027	10/28/19
B 3.0-12	81	DRR 19-1027	10/28/19
B 3.0-13	81	DRR 19-1027	10/28/19
B 3.0-14	81	DRR 19-1027	10/28/19
B 3.0-15	81	DRR 19-1027	10/28/19
B 3.0-16	81	DRR 19-1027	10/28/19
B 3.0-17	81	DRR 19-1027	10/28/19
TAB – B 3.1 REACTIVITY CONTROL SYSTEMS			
B 3.1.1-1	0	Amend. No. 123	12/18/99
B 3.1.1-2	0	Amend. No. 123	12/18/99
B 3.1.1-3	95	N/A	4/25/24
B 3.1.1-4	81	DRR 19-1027	10/28/19
B 3.1.1-5	89	DRR 21-0966	7/7/21
B 3.1.2-1	0	Amend. No. 123	12/18/99
B 3.1.2-2	0	Amend. No. 123	12/18/99
B 3.1.2-3	0	Amend. No. 123	12/18/99
B 3.1.2-4	0	Amend. No. 123	12/18/99
B 3.1.2-5	89	DRR 21-0966	7/7/21
B 3.1.3-1	96	N/A	12/11/2024
B 3.1.3-2	96	N/A	12/11/2024
B 3.1.3-3	96	N/A	12/11/2024

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TAB – B 3.1 REACTIVITY CONTROL SYSTEMS (continued)			
B 3.1.3-4	96	N/A	12/11/2024
B 3.1.3-5	96	N/A	12/11/2024
B 3.1.3-6	95	N/A	4/25/24
B 3.1.4-1	0	Amend. No. 123	12/18/99
B 3.1.4-2	0	Amend. No. 123	12/18/99
B 3.1.4-3	48	DRR 10-3740	12/28/10
B 3.1.4-4	0	Amend. No. 123	12/18/99
B 3.1.4-5	0	Amend. No. 123	12/18/99
B 3.1.4-6	48	DRR 10-3740	12/28/10
B 3.1.4-7	0	Amend. No. 123	12/18/99
B 3.1.4-8	89	DRR 21-0966	7/7/21
B 3.1.4-9	89	DRR 21-0966	7/7/21
B 3.1.5-1	0	Amend. No. 123	12/18/99
B 3.1.5-2	0	Amend. No. 123	12/18/99
B 3.1.5-3	0	Amend. No. 123	12/18/99
B 3.1.5-4	89	DRR 21-0966	7/7/21
B 3.1.6-1	0	Amend. No. 123	12/18/99
B 3.1.6-2	0	Amend. No. 123	12/18/99
B 3.1.6-3	0	Amend. No. 123	12/18/99
B 3.1.6-4	0	Amend. No. 123	12/18/99
B 3.1.6-5	89	DRR 21-0966	7/7/21
B 3.1.6-6	95	N/A	4/25/24
B 3.1.7-1	0	Amend. No. 123	12/18/99
B 3.1.7-2	0	Amend. No. 123	12/18/99
B 3.1.7-3	48	DRR 10-3740	12/28/10
B 3.1.7-4	48	DRR 10-3740	12/28/10
B 3.1.7-5	48	DRR 10-3740	12/28/10
B 3.1.7-6	0	Amend. No. 123	12/18/99
B 3.1.8-1	0	Amend. No. 123	12/18/99
B 3.1.8-2	0	Amend. No. 123	12/18/99
B 3.1.8-3	15	DRR 03-0860	7/10/03
B 3.1.8-4	15	DRR 03-0860	7/10/03
B 3.1.8-5	89	DRR 21-0966	7/7/21
B 3.1.8-6	89	DRR 21-0966	7/7/21
B 3.1.9-1	84	DRR 20-0400	08/18/20
B 3.1.9-2	84	DRR 20-0400	08/18/20
B 3.1.9-3	84	DRR 20-0400	08/18/20
B 3.1.9-4	84	DRR 20-0400	08/18/20
B 3.1.9-5	89	DRR 21-0966	7/7/21
TAB – B 3.2 POWER DISTRIBUTION LIMITS			
B 3.2.1-1	48	DRR 10-3740	12/28/10
B 3.2.1-2	0	Amend. No. 123	12/18/99
B 3.2.1-3	48	DRR 10-3740	12/28/10
B 3.2.1-4	48	DRR 10-3740	12/28/10
B 3.2.1-5	48	DRR 10-3740	12/28/10
B 3.2.1-6	48	DRR 10-3740	12/28/10
B 3.2.1-7	48	DRR 10-3740	12/28/10
B 3.2.1-8	89	DRR 21-0966	7/7/21
B 3.2.1-9	89	DRR 21-0966	7/7/21
B 3.2.1-10	70	DRR 15-0944	4/28/15

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TAB – B 3.2 POWER DISTRIBUTION LIMITS (continued)			
B 3.2.2-1	48	DRR 10-3740	12/28/10
B 3.2.2-2	0	Amend. No. 123	12/18/99
B 3.2.2-3	48	DRR 10-3740	12/28/10
B 3.2.2-4	48	DRR 10-3740	12/28/10
B 3.2.2-5	48	DRR 10-3740	12/28/10
B 3.2.2-6	89	DRR 21-0966	7/7/21
B 3.2.3-1	0	Amend. No. 123	12/18/99
B 3.2.3-2	0	Amend. No. 123	12/18/99
B 3.2.3-3	89	DRR 21-0966	7/7/21
B 3.2.4-1	0	Amend. No. 123	12/18/99
B 3.2.4-2	0	Amend. No. 123	12/18/99
B 3.2.4-3	48	DRR 10-3740	12/28/10
B 3.2.4-4	0	Amend. No. 123	12/18/99
B 3.2.4-5	48	DRR 10-3740	12/28/10
B 3.2.4-6	89	DRR 21-0966	7/7/21
B 3.2.4-7	89	DRR 21-0966	7/7/21
TAB – B 3.3 INSTRUMENTATION			
B 3.3.1-1	84	DRR 20-0400	08/18/20
B 3.3.1-2	0	Amend. No. 123	12/18/99
B 3.3.1-3	0	Amend. No. 123	12/18/99
B 3.3.1-4	0	Amend. No. 123	12/18/99
B 3.3.1-5	0	Amend. No. 123	12/18/99
B 3.3.1-6	0	Amend. No. 123	12/18/99
B 3.3.1-7	5	DRR 00-1427	10/12/00
B 3.3.1-8	0	Amend. No. 123	12/18/99
B 3.3.1-9	84	DRR 20-0400	08/18/20
B 3.3.1-10	84	DRR 20-0400	08/18/20
B 3.3.1-11	95	N/A	4/25/24
B 3.3.1-12	95	N/A	4/25/24
B 3.3.1-13	95	N/A	4/25/24
B 3.3.1-14	95	N/A	4/25/24
B 3.3.1-15	95	N/A	4/25/24
B 3.3.1-16	95	N/A	4/25/24
B 3.3.1-17	95	N/A	4/25/24
B 3.3.1-18	95	N/A	4/25/24
B 3.3.1-19	95	N/A	4/25/24
B 3.3.1-20	84	DRR 20-0400	08/18/20
B 3.3.1-21	84	DRR 20-0400	08/18/20
B 3.3.1-22	84	DRR 20-0400	08/18/20
B 3.3.1-23	84	DRR 20-0400	08/18/20
B 3.3.1-24	84	DRR 20-0400	08/18/20
B 3.3.1-25	84	DRR 20-0400	08/18/20
B 3.3.1-26	84	DRR 20-0400	08/18/20
B 3.3.1-27	84	DRR 20-0400	08/18/20
B 3.3.1-28	84	DRR 20-0400	08/18/20
B 3.3.1-29	84	DRR 20-0400	08/18/20
B 3.3.1-30	84	DRR 20-0400	08/18/20
B 3.3.1-31	84	DRR 20-0400	08/18/20
B 3.3.1-32	84	DRR 20-0400	08/18/20
B 3.3.1-33	84	DRR 20-0400	08/18/20

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TAB – B 3.3 INSTRUMENTATION (continued)			
B 3.3.1-34	84	DRR 20-0400	08/18/20
B 3.3.1-35	84	DRR 20-0400	08/18/20
B 3.3.1-36	95	N/A	4/25/24
B 3.3.1-37	84	DRR 20-0400	08/18/20
B 3.3.1-38	84	DRR 20-0400	08/18/20
B 3.3.1-39	84	DRR 20-0400	08/18/20
B 3.3.1-40	84	DRR 20-0400	08/18/20
B 3.3.1-41	84	DRR 20-0400	08/18/20
B 3.3.1-42	84	DRR 20-0400	08/18/20
B 3.3.1-43	84	DRR 20-0400	08/18/20
B 3.3.1-44	84	DRR 20-0400	08/18/20
B 3.3.1-45	84	DRR 20-0400	08/18/20
B 3.3.1-46	84	DRR 20-0400	08/18/20
B 3.3.1-47	89	DRR 21-0966	7/7/21
B 3.3.1-48	84	DRR 20-0400	08/18/20
B 3.3.1-49	89	DRR 21-0966	7/7/21
B 3.3.1-50	89	DRR 21-0966	7/7/21
B 3.3.1-51	89	DRR 21-0966	7/7/21
B 3.3.1-52	89	DRR 21-0966	7/7/21
B 3.3.1-53	89	DRR 21-0966	7/7/21
B 3.3.1-54	89	DRR 21-0966	7/7/21
B 3.3.1-55	89	DRR 21-0966	7/7/21
B 3.3.1-56	97	N/A	5/27/25
B 3.3.1-57	97	N/A	5/27/25
B 3.3.1-58	97	N/A	5/27/25
B 3.3.1-59	97	N/A	5/27/25
B 3.3.1-60	97	N/A	5/27/25
B 3.3.1-61	97	N/A	5/27/25
B 3.3.1-62 (new)	97	N/A	5/27/25
B 3.3.2-1	84	DRR 20-0400	08/18/20
B 3.3.2-2	0	Amend. No. 123	12/18/99
B 3.3.2-3	0	Amend. No. 123	12/18/99
B 3.3.2-4	0	Amend. No. 123	12/18/99
B 3.3.2-5	0	Amend. No. 123	12/18/99
B 3.3.2-6	7	DRR 01-0474	5/1/01
B 3.3.2-7	0	Amend. No. 123	12/18/99
B 3.3.2-8	0	Amend. No. 123	12/18/99
B 3.3.2-9	0	Amend. No. 123	12/18/99
B 3.3.2-10	0	Amend. No. 123	12/18/99
B 3.3.2-11	0	Amend. No. 123	12/18/99
B 3.3.2-12	81	DRR 19-1027	10/28/19
B 3.3.2-13	0	Amend. No. 123	12/18/99
B 3.3.2-14	2	DRR 00-0147	4/24/00
B 3.3.2-15	0	Amend. No. 123	12/18/99
B 3.3.2-16	0	Amend. No. 123	12/18/99
B 3.3.2-17	0	Amend. No. 123	12/18/99
B 3.3.2-18	0	Amend. No. 123	12/18/99
B 3.3.2-19	37	DRR 08-0503	4/8/08
B 3.3.2-20	37	DRR 08-0503	4/8/08
B 3.3.2-21	37	DRR 08-0503	4/8/08
B 3.3.2-23	37	DRR 08-0503	4/8/08

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B 3.3.2-24	39	DRR 08-1096	8/28/08
B 3.3.2-25	39	DRR 08-1096	8/28/08
B 3.3.2-26	39	DRR 08-1096	8/28/08
B 3.3.2-27	37	DRR 08-0503	4/8/08
B 3.3.2-28	84	DRR 20-0400	08/18/20
B 3.3.2-29	0	Amend. No. 123	12/18/99
B 3.3.2-30	0	Amend. No. 123	12/18/99
B 3.3.2-31	52	DRR 11-0724	4/11/11
B 3.3.2-32	52	DRR 11-0724	4/11/11
B 3.3.2-33	0	Amend. No. 123	12/18/99
B 3.3.2-34	0	Amend. No. 123	12/18/99
B 3.3.2-35	20	DRR 04-1533	2/16/05
B 3.3.2-36	20	DRR 04-1533	2/16/05
B 3.3.2-37	20	DRR 04-1533	2/16/05
B 3.3.2-38	20	DRR 04-1533	2/16/05
B 3.3.2-39	25	DRR 06-0800	5/18/06
B 3.3.2-40	20	DRR 04-1533	2/16/05
B 3.3.2-41	45	Amend. No. 187 (ETS)	3/5/10
B 3.3.2-42	45	Amend. No. 187 (ETS)	3/5/10
B 3.3.2-43	20	DRR 04-1533	2/16/05
B 3.3.2-44	91	DRR 22-0243	3/2/22
B 3.3.2-45	92	DRR 22-0767	11/3/22
B 3.3.2-46	89	DRR 21-0966	7/7/21
B 3.3.2-47	89	DRR 21-0966	7/7/21
B 3.3.2-48	89	DRR 21-0966	7/7/21
B 3.3.2-49	89	DRR 21-0966	7/7/21
B 3.3.2-50	97	N/A	5/27/25
B 3.3.2-51	97	N/A	5/27/25
B 3.3.2-52	97	N/A	5/27/25
B 3.3.2-53	43	DRR 09-1416	9/2/09
B 3.3.2-54	43	DRR 09-1416	9/2/09
B 3.3.2-55	43	DRR 09-1416	9/2/09
B 3.3.2-56	43	DRR 09-1416	9/2/09
B 3.3.2-57	43	DRR 09-1416	9/2/09
B 3.3.3-1	0	Amend. No. 123	12/18/99
B 3.3.3-2	5	DRR 00-1427	10/12/00
B 3.3.3-3	0	Amend. No. 123	12/18/99
B 3.3.3-4	0	Amend. No. 123	12/18/99
B 3.3.3-5	0	Amend. No. 123	12/18/99
B 3.3.3-6	8	DRR 01-1235	9/19/01
B 3.3.3-7	21	DRR 05-0707	4/20/05
B 3.3.3-8	81	DRR 19-1027	10/28/19
B 3.3.3-9	8	DRR 01-1235	9/19/01
B 3.3.3-10	19	DRR 04-1414	10/12/04
B 3.3.3-11	19	DRR 04-1414	10/12/04
B 3.3.3-12	21	DRR 05-0707	4/20/05
B 3.3.3-13	89	DRR 21-0966	7/7/21
B 3.3.3-14	89	DRR 21-0966	7/7/21
B 3.3.3-15	8	DRR 01-1235	9/19/01
B 3.3.4-1	0	Amend. No. 123	12/18/99
B 3.3.4-2	9	DRR 02-1023	2/28/02

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TAB – B 3.3 INSTRUMENTATION (continued)			
B 3.3.4-3	15	DRR 03-0860	7/10/03
B 3.3.4-4	19	DRR 04-1414	10/12/04
B 3.3.4-5	89	DRR 21-0966	7/7/21
B 3.3.4-6	89	DRR 21-0966	7/7/21
B 3.3.5-1	88	DRR 21-0591	4/28/21
B 3.3.5-2	88	DRR 21-0591	4/28/21
B 3.3.5-3	1	DRR 99-1624	12/18/99
B 3.3.5-4	1	DRR 99-1624	12/18/99
B 3.3.5-5	0	Amend. No. 123	12/18/99
B 3.3.5-6	89	DRR 21-0966	7/7/21
B 3.3.5-7	89	DRR 21-0966	7/7/21
B 3.3.6-1	81	DRR 19-1027	10/28/19
B 3.3.6-2	81	DRR 19-1027	10/28/19
B 3.3.6-3	0	Amend. No. 123	12/18/99
B 3.3.6-4	0	Amend. No. 123	12/18/99
B 3.3.6-5	89	DRR 21-0966	7/7/21
B 3.3.6-6	89	DRR 21-0966	7/7/21
B 3.3.6-7	89	DRR 21-0966	7/7/21
B 3.3.7-1	81	DRR 19-1027	10/28/19
B 3.3.7-2	81	DRR 19-1027	10/28/19
B 3.3.7-3	57	DRR 13-0006	1/16/13
B 3.3.7-4	0	Amend. No. 123	12/18/99
B 3.3.7-5	0	Amend. No. 123	12/18/99
B 3.3.7-6	89	DRR 21-0966	7/7/21
B 3.3.7-7	89	DRR 21-0966	7/7/21
B 3.3.7-8	89	DRR 21-0966	7/7/21
B 3.3.8-1	84	DRR 20-0400	8/18/20
B 3.3.8-2	0	Amend. No. 123	12/18/99
B 3.3.8-3	57	DRR 13-0006	1/16/13
B 3.3.8-4	57	DRR 13-0006	1/16/13
B 3.3.8-5	89	DRR 21-0966	7/7/21
B 3.3.8-6	89	DRR 21-0966	7/7/21
B 3.3.8-7	89	DRR 21-0966	7/7/21
TAB – B 3.4 REACTOR COOLANT SYSTEM (RCS)			
B 3.4.1-1	84	DRR 20-0400	08/18/20
B 3.4.1-2	84	DRR 20-0400	08/18/20
B 3.4.1-3	10	DRR 02-0411	4/5/02
B 3.4.1-4	0	Amend. No. 123	12/18/99
B 3.4.1-5	89	DRR 21-0966	7/7/21
B 3.4.1-6	89	DRR 21-0966	7/7/21
B 3.4.2-1	0	Amend. No. 123	12/18/99
B 3.4.2-2	0	Amend. No. 123	12/18/99
B 3.4.2-3	89	DRR 21-0966	7/7/21
B 3.4.3-1	67	DRR 15-0116	2/10/15
B 3.4.3-2	0	Amend. No. 123	12/18/99
B 3.4.3-3	0	Amend. No. 123	12/18/99
B 3.4.3-4	0	Amend. No. 123	12/18/99
B 3.4.3-5	0	Amend. No. 123	12/18/99
B 3.4.3-6	89	DRR 21-0966	7/7/21
B 3.4.3-7	0	Amend. No. 123	12/18/99

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TAB – B 3.4 REACTOR COOLANT SYSTEM (RCS) (continued)			
B 3.4.4-1	0	Amend. No. 123	12/18/99
B 3.4.4-2	29	DRR 06-1984	10/17/06
B 3.4.4-3	89	DRR 21-0966	7/7/21
B 3.4.5-1	0	Amend. No. 123	12/18/99
B 3.4.5-2	53	DRR 11-1513	7/18/11
B 3.4.5-3	29	DRR 06-1984	10/17/06
B 3.4.5-4	0	Amend. No. 123	12/18/99
B 3.4.5-5	89	DRR 21-0966	7/7/21
B 3.4.5-6	89	DRR 21-0966	7/7/21
B 3.4.6-1	53	DRR 11-1513	7/18/11
B 3.4.6-2	72	DRR 15-1918	10/26/15
B 3.4.6-3	12	DRR 02-1062	9/26/02
B 3.4.6-4	89	DRR 21-0966	7/7/21
B 3.4.6-5	75	DRR 16-1909	10/26/16
B 3.4.6-6	89	DRR 21-0966	7/7/21
B 3.4.7-1	12	DRR 02-1062	9/26/02
B 3.4.7-2	17	DRR 04-0453	5/26/04
B 3.4.7-3	90	DRR 21-1229	11/3/21
B 3.4.7-4	89	DRR 21-0966	7/7/21
B 3.4.7-5	89	DRR 21-0966	7/7/21
B 3.4.7-6	89	DRR 21-0966	7/7/21
B 3.4.8-1	53	DRR 11-1513	7/18/11
B 3.4.8-2	90	DRR 21-1229	11/3/21
B 3.4.8-3	89	DRR 21-0966	7/7/21
B 3.4.8-4	89	DRR 21-0966	7/7/21
B 3.4.8-5	89	DRR 21-0966	7/7/21
B 3.4.9-1	0	Amend. No. 123	12/18/99
B 3.4.9-2	0	Amend. No. 123	12/18/99
B 3.4.9-3	0	Amend. No. 123	12/18/99
B 3.4.9-4	89	DRR 21-0966	7/7/21
B 3.4.10-1	5	DRR 00-1427	10/12/00
B 3.4.10-2	5	DRR 00-1427	10/12/00
B 3.4.10-3	0	Amend. No. 123	12/18/99
B 3.4.10-4	32	DRR 07-0139	2/7/07
B 3.4.11-1	0	Amend. No. 123	12/18/99
B 3.4.11-2	1	DRR 99-1624	12/18/99
B 3.4.11-3	19	DRR 04-1414	10/12/04
B 3.4.11-4	0	Amend. No. 123	12/18/99
B 3.4.11-5	1	DRR 99-1624	12/18/99
B 3.4.11-6	0	Amend. No. 123	12/18/99
B 3.4.11-7	89	DRR 21-0966	7/7/21
B 3.4.12-1	61	DRR 14-0346	2/27/14
B 3.4.12-2	61	DRR 14-0346	2/27/14
B 3.4.12-3	0	Amend. No. 123	12/18/99
B 3.4.12-4	61	DRR 14-0346	2/27/14
B 3.4.12-5	61	DRR 14-0346	2/27/14
B 3.4.12-6	56	DRR 12-1792	11/7/12
B 3.4.12-7	61	DRR 14-0346	2/27/14
B 3.4.12-8	1	DRR 99-1624	12/18/99
B 3.4.12-9	56	DRR 12-1792	11/7/12
B 3.4.12-10	0	Amend. No. 123	12/18/99

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TAB – B 3.4 REACTOR COOLANT SYSTEM (RCS) (continued)			
B 3.4.12-11	61	DRR 14-0346	2/27/14
B 3.4.12-12	89	DRR 21-0966	7/7/21
B 3.4.12-13	89	DRR 21-0966	7/7/21
B 3.4.12-14	89	DRR 21-0966	7/7/21
B 3.4.13-1	0	Amend. No. 123	12/18/99
B 3.4.13-2	94	N/A	10/6/23
B 3.4.13-3	94	N/A	10/6/23
B 3.4.13-4	94	N/A	10/6/23
B 3.4.13-5	94	N/A	10/6/23
B 3.4.13-6	89	DRR 21-0966	7/7/21
B 3.4.14-1	0	Amend. No. 123	12/18/99
B 3.4.14-2	0	Amend. No. 123	12/18/99
B 3.4.14-3	0	Amend. No. 123	12/18/99
B 3.4.14-4	0	Amend. No. 123	12/18/99
B 3.4.14-5	89	DRR 21-0966	7/7/21
B 3.4.14-6	89	DRR 21-0966	7/7/21
B 3.4.15-1	31	DRR 06-2494	12/13/06
B 3.4.15-2	31	DRR 06-2494	12/13/06
B 3.4.15-3	33	DRR 07-0656	5/1/07
B 3.4.15-4	33	DRR 07-0656	5/1/07
B 3.4.15-5	65	DRR 14-2146	9/30/14
B 3.4.15-6	31	DRR 06-2494	12/13/06
B 3.4.15-7	89	DRR 21-0966	7/7/21
B 3.4.15-8	31	DRR 06-2494	12/13/06
B 3.4.16-1	82	DRR 20-0077	1/29/20
B 3.4.16-2	84	DRR 20-0400	08/18/20
B 3.4.16-3	82	DRR 20-0077	1/29/20
B 3.4.16-4	92	DRR 22-0767	11/3/22
B 3.4.16-5	92	DRR 22-0767	11/3/22
B 3.4.17-1	29	DRR 06-1984	10/17/06
B 3.4.17-2	81	DRR 19-1027	10/28/19
B 3.4.17-3	29	DRR 06-1984	10/17/06
B 3.4.17-4	81	DRR 19-1027	10/28/19
B 3.4.17-5	57	DRR 13-0006	1/16/13
B 3.4.17-6	57	DRR 13-0006	1/16/13
B 3.4.17-7	81	DRR 19-1027	10/28/19
TAB – B 3.5 EMERGENCY CORE COOLING SYSTEMS (ECCS)			
B 3.5.1-1	0	Amend. No. 123	12/18/99
B 3.5.1-2	0	Amend. No. 123	12/18/99
B 3.5.1-3	73	DRR 15-2135	11/17/15
B 3.5.1-4	73	DRR 15-2135	11/17/15
B 3.5.1-5	1	DRR 99-1624	12/18/99
B 3.5.1-6	89	DRR 21-0966	7/7/21
B 3.5.1-7	89	DRR 21-0966	7/7/21
B 3.5.1-8	1	DRR 99-1624	12/18/99
B 3.5.2-1	84	DRR 20-0400	08/18/20
B 3.5.2-2	0	Amend. No. 123	12/18/99
B 3.5.2-3	0	Amend. No. 123	12/18/99
B 3.5.2-4	0	Amend. No. 123	12/18/99
B 3.5.2-5	72	DRR 15-1918	10/26/15

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TAB – B 3.5 EMERGENCY CORE COOLING SYSTEMS (ECCS) (continued)			
B 3.5.2-6	42	DRR 09-1009	7/16/09
B 3.5.2-7	89	DRR 21-0966	7/7/21
B 3.5.2-8	89	DRR 21-0966	7/7/21
B 3.5.2-9	89	DRR 21-0966	7/7/21
B 3.5.2-10	89	DRR 21-0966	7/7/21
B 3.5.2-11	89	DRR 21-0966	7/7/21
B 3.5.2-12	72	DRR 15-1918	10/26/15
B 3.5.3-1	56	DRR 12-1792	11/7/12
B 3.5.3-2	72	DRR 15-1918	10/26/15
B 3.5.3-3	56	DRR 12-1792	11/7/12
B 3.5.3-4	56	DRR 12-1792	11/7/12
B 3.5.4-1	0	Amend. No. 123	12/18/99
B 3.5.4-2	0	Amend. No. 123	12/18/99
B 3.5.4-3	0	Amend. No. 123	12/18/99
B 3.5.4-4	0	Amend. No. 123	12/18/99
B 3.5.4-5	89	DRR 21-0966	7/7/21
B 3.5.4-6	89	DRR 21-0966	7/7/21
B 3.5.5-1	21	DRR 05-0707	4/20/05
B 3.5.5-2	21	DRR 05-0707	4/20/05
B 3.5.5-3	89	DRR 21-0966	7/7/21
B 3.5.5-4	21	DRR 05-0707	4/20/05
TAB – B 3.6 CONTAINMENT SYSTEMS			
B 3.6.1-1	0	Amend. No. 123	12/18/99
B 3.6.1-2	81	DRR 19-1027	10/28/19
B 3.6.1-3	0	Amend. No. 123	12/18/99
B 3.6.1-4	96	N/A	12/11/2024
B 3.6.1-5	96	N/A	12/11/2024
B 3.6.2-1	81	DRR 19-1027	10/28/19
B 3.6.2-2	0	Amend. No. 123	12/18/99
B 3.6.2-3	0	Amend. No. 123	12/18/99
B 3.6.2-4	0	Amend. No. 123	12/18/99
B 3.6.2-5	0	Amend. No. 123	12/18/99
B 3.6.2-6	89	DRR 21-0966	7/7/21
B 3.6.2-7	89	DRR 21-0966	7/7/21
B 3.6.3-1	0	Amend. No. 123	12/18/99
B 3.6.3-2	84	DRR 20-0400	08/18/20
B 3.6.3-3	90	DRR 21-1229	11/3/21
B 3.6.3-4	49	DRR 11-0014	1/31/11
B 3.6.3-5	49	DRR 11-0014	1/31/11
B 3.6.3-6	49	DRR 11-0014	1/31/11
B 3.6.3-7	90	DRR 21-1229	11/3/21
B 3.6.3-8	36	DRR 08-0255	3/11/08
B 3.6.3-9	90	DRR 21-1299	11/3/21
B 3.6.3-10	89	DRR 21-0966	7/7/21
B 3.6.3-11	36	DRR 08-0255	3/11/08
B 3.6.3-12	89	DRR 21-0966	7/7/21
B 3.6.3-13	89	DRR 21-0966	7/7/21
B 3.6.3-14	36	DRR 08-0255	3/11/08
B 3.6.3-15	39	DRR 08-1096	8/28/08
B 3.6.3-16	39	DRR 08-1096	8/28/08

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TAB – B 3.6 CONTAINMENT SYSTEMS (continued)			
B 3.6.3-17	36	DRR 08-0255	3/11/08
B 3.6.3-18	36	DRR 08-0255	3/11/08
B 3.6.3-19	82	DRR 20-0077	1/29/20
B 3.6.4-1	39	DRR 08-1096	8/28/08
B 3.6.4-2	0	Amend. No. 123	12/18/99
B 3.6.4-3	89	DRR 21-0966	7/7/21
B 3.6.5-1	0	Amend. No. 123	12/18/99
B 3.6.5-2	37	DRR 08-0503	4/8/08
B 3.6.5-3	89	DRR 21-0966	7/7/21
B 3.6.5-4	0	Amend. No. 123	12/18/99
B 3.6.6-1	81	DRR 19-1027	10/28/19
B 3.6.6-2	63	DRR 14-1572	7/1/14
B 3.6.6-3	37	DRR 08-0503	4/8/08
B 3.6.6-4	81	DRR 19-1027	10/28/19
B 3.6.6-5	0	Amend. No. 123	12/18/99
B 3.6.6-6	18	DRR 04-1018	9/1/04
B 3.6.6-7	89	DRR 21-0966	7/7/21
B 3.6.6-8	89	DRR 21-0966	7/7/21
B 3.6.6-9	72	DRR 15-1918	10/26/15
B 3.6.6-10	89	DRR 21-0966	7/7/21
B 3.6.6.11	80	DRR 19-0524	5/30/19
B 3.6.7-1	0	Amend. No. 123	12/18/99
B 3.6.7-2	81	DRR 19-1027	10/28/19
B 3.6.7-3	89	DRR 21-0966	7/7/21
B 3.6.7-4	89	DRR 21-0966	7/7/21
B 3.6.7-5	89	DRR 21-0966	7/7/21
TAB – B 3.7 PLANT SYSTEMS			
B 3.7.1-1	0	Amend. No. 123	12/18/99
B 3.7.1-2	84	DRR 20-0400	08/18/20
B 3.7.1-3	0	Amend. No. 123	12/18/99
B 3.7.1-4	84	DRR 20-0400	08/18/20
B 3.7.1-5	84	DRR 20-0400	08/18/20
B 3.7.1-6	84	DRR 20-0400	08/18/20
B 3.7.2-1	44	DRR 09-1744	10/28/09
B 3.7.2-2	82	DRR 20-0077	1/29/20
B 3.7.2-3	82	DRR 20-0077	1/29/20
B 3.7.2-4	81	DRR 19-1027	10/28/19
B 3.7.2-5	82	DRR 20-0077	1/29/20
B 3.7.2-6	82	DRR 20-0077	1/29/20
B 3.7.2-7	82	DRR 20-0077	1/29/20
B 3.7.2-8	82	DRR 20-0077	1/29/20
B 3.7.2-9	89	DRR 21-0966	7/7/21
B 3.7.2-10	81	DRR 19-1027	10/28/19
B 3.7.2-11	44	DRR 09-1744	10/28/09
B 3.7.3-1	37	DRR 08-0503	4/8/08
B 3.7.3-2	50	DRR 11-0449	3/9/11
B 3.7.3-3	37	DRR 08-0503	4/8/08
B 3.7.3-4	37	DRR 08-0503	4/8/08
B 3.7.3-5	37	DRR 08-0503	4/8/08
B 3.7.3-6	37	DRR 08-0503	4/8/08

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TAB – B 3.7 PLANT SYSTEMS (continued)			
B 3.7.3-7	37	DRR 08-0503	4/8/08
B 3.7.3-8	37	DRR 08-0503	4/8/08
B 3.7.3-9	66	DRR 14-2329	11/6/14
B 3.7.3-10	89	DRR 21-0966	7/7/21
B 3.7.3-11	37	DRR 08-0503	4/8/08
B 3.7.4-1	1	DRR 99-1624	12/18/99
B 3.7.4-2	84	DRR 20-0400	08/18/20
B 3.7.4-3	19	DRR 04-1414	10/12/04
B 3.7.4-4	19	DRR 04-1414	10/12/04
B 3.7.4-5	89	DRR 21-0966	7/7/21
B 3.7.5-1	54	DRR 11-2394	11/16/11
B 3.7.5-2	54	DRR 11-2394	11/16/11
B 3.7.5-3	0	Amend. No. 123	12/18/99
B 3.7.5-4	85	DRR 20-0988	10/24/20
B 3.7.5-5	76	DRR 17-0343	2/21/17
B 3.7.5-6	85	DRR 20-0988	10/24/20
B 3.7.5-7	90	DRR 21-1229	11/3/21
B 3.7.5-8	90	DRR 21-1229	11/3/21
B 3.7.5-9	90	DRR 21-1229	11/3/21
B 3.7.5-10	90	DRR 21-1229	11/3/21
B 3.7.6-1	0	Amend. No. 123	12/18/99
B 3.7.6-2	0	Amend. No. 123	12/18/99
B 3.7.6-3	89	DRR 21-0966	7/7/21
B 3.7.7-1	0	Amend. No. 123	12/18/99
B 3.7.7-2	77	DRR 17-1001	6/22/17
B 3.7.7-3	92	DRR 22-0767	11/3/22
B 3.7.7-4	89	DRR 21-0966	7/7/21
B 3.7.8-1	0	Amend. No. 123	12/18/99
B 3.7.8-2	0	Amend. No. 123	12/18/99
B 3.7.8-3	0	Amend. No. 123	12/18/99
B 3.7.8-4	89	DRR 21-0966	7/7/21
B 3.7.8-5	89	DRR 21-0966	7/7/21
B 3.7.9-1	3	Amend. No. 134	7/14/00
B 3.7.9-2	3	Amend. No. 134	7/14/00
B 3.7.9-3	89	DRR 21-0966	7/7/21
B 3.7.9-4	3	Amend. No. 134	7/14/00
B 3.7.10-1	64	DRR 14-1822	8/28/14
B 3.7.10-2	81	DRR 19-1027	10/28/19
B 3.7.10-3	81	DRR 19-1027	10/28/19
B 3.7.10-4	81	DRR 19-1027	10/28/19
B 3.7.10-5	81	DRR 19-1027	10/28/19
B 3.7.10-6	57	DRR 13-0006	1/16/13
B 3.7.10-7	89	DRR 21-0966	7/7/21
B 3.7.10-8	89	DRR 21-0966	7/7/21
B 3.7.10-9	81	DRR 19-1027	10/28/19
B 3.7.11-1	0	Amend. No. 123	12/18/99
B 3.7.11-2	57	DRR 13-0006	1/16/13
B 3.7.11-3	89	DRR 21-0966	7/7/21
B 3.7.11-4	63	DRR 14-1572	7/1/14
B 3.7.12-1	0	Amend. No. 123	12/18/99
B 3.7.13-1	24	DRR 06-0051	2/28/06

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TAB – B 3.7 PLANT SYSTEMS (continued)			
B 3.7.13-2	81	DRR 19-1027	10/28/19
B 3.7.13-3	81	DRR 19-1027	10/28/19
B 3.7.13-4	81	DRR 19-1027	10/28/19
B 3.7.13-5	96	N/A	12/11/2024
B 3.7.13-6	89	DRR 21-0966	7/7/21
B 3.7.13-7	89	DRR 21-0966	7/7/21
B 3.7.13-8	89	DRR 21-0966	7/7/21
B 3.7.14-1	0	Amend. No. 123	12/18/99
B 3.7.15-1	81	DRR 19-1027	10/28/19
B 3.7.15-2	89	DRR 21-0966	7/7/21
B 3.7.15-3	81	DRR 19-1027	10/28/19
B 3.7.16-1	5	DRR 00-1427	10/12/00
B 3.7.16-2	95	N/A	4/25/24
B 3.7.16-3	89	DRR 21-0966	7/7/21
B 3.7.17-1	7	DRR 01-0474	5/1/01
B 3.7.17-2	7	DRR 01-0474	5/1/01
B 3.7.17-3	5	DRR 00-1427	10/12/00
B 3.7.18-1	81	DRR 19-1027	10/28/19
B 3.7.18-2	81	DRR 19-1027	10/28/19
B 3.7.18-3	89	DRR 21-0966	7/7/21
B 3.7.19-1	44	DRR 09-1744	10/28/09
B 3.7.19-2	54	DRR 11-2394	11/16/11
B 3.7.19-3	54	DRR 11-2394	11/16/11
B 3.7.19-4	61	DRR 14-0346	2/27/14
B 3.7.19-5	61	DRR 14-0346	2/27/14
B 3.7.19-6	89	DRR 21-0966	7/7/21
B 3.7.19-7	89	DRR 21-0966	7/7/21
B 3.7.20-1	79	DRR 18-1579	10/22/18
B 3.7.20-2	90	DRR 21-1229	11/3/21
B 3.7.20-3	85	DRR 20-0988	10/24/20
B 3.7.20-4	89	DRR 21-0966	7/7/21
B 3.7.20-5	89	DRR 21-0966	7/7/21
TAB – B 3.8 ELECTRICAL POWER SYSTEMS			
B 3.8.1-1	88	DRR 21-0591	4/28/21
B 3.8.1-2	93	DRR 22-1363	2/2/23
B 3.8.1-3	95	N/A	4/25/24
B 3.8.1-4	96	N/A	12/11/2024
B 3.8.1-5	93	DRR 22-1363	2/2/23
B 3.8.1-6	93	DRR 22-1363	2/2/23
B 3.8.1-7	93	DRR 22-1363	2/2/23
B 3.8.1-8	93	DRR 22-1363	2/2/23
B 3.8.1-9	93	DRR 22-1363	2/2/23
B 3.8.1-10	93	DRR 22-1363	2/2/23
B 3.8.1-11	93	DRR 22-1363	2/2/23
B 3.8.1-12	93	DRR 22-1363	2/2/23
B 3.8.1-13	93	DRR 22-1363	2/2/23
B 3.8.1-14	93	DRR 22-1363	2/2/23
B 3.8.1-15	47	DRR 10-1089	6/16/10
B 3.8.1-16	26	DRR 06-1350	7/24/06
B 3.8.1-17	93	DRR 22-1363	2/2/23

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TAB – B 3.8 ELECTRICAL POWER SYSTEMS (continued)			
B 3.8.1-18	95	N/A	4/25/24
B 3.8.1-19	89	DRR 21-0966	7/7/21
B 3.8.1-20	89	DRR 21-0966	7/7/21
B 3.8.1-21	93	DRR 22-1363	2/2/23
B 3.8.1-22	89	DRR 21-0966	7/7/21
B 3.8.1-23	93	DRR 22-1363	2/2/23
B 3.8.1-24	74	DRR 16-1182	7/7/16
B 3.8.1-25	89	DRR 21-0966	7/7/21
B 3.8.1-26	89	DRR 21-0966	7/7/21
B 3.8.1-27	89	DRR 21-0966	7/7/21
B 3.8.1-28	93	DRR 22-1363	2/2/23
B 3.8.1-29	96	N/A	12/11/2024
B 3.8.1-30	96	N/A	12/11/2024
B 3.8.1-31	89	DRR 21-0966	7/7/21
B 3.8.1-32	89	DRR 21-0966	7/7/21
B 3.8.1-33	89	DRR 21-0966	7/7/21
B 3.8.1-34	93	DRR 22-1363	2/2/23
B 3.8.2-1	57	DRR 13-0006	1/16/13
B 3.8.2-2	0	Amend. No. 123	12/18/99
B 3.8.2-3	92	DRR 22-0767	11/3/22
B 3.8.2-4	92	DRR 22-0767	11/3/22
B 3.8.2-5	57	DRR 13-0006	1/16/13
B 3.8.2-6	57	DRR 13-0006	1/16/13
B 3.8.2-7	57	DRR 13-0006	1/16/13
B 3.8.3-1	1	DRR 99-1624	12/18/99
B 3.8.3-2	90	DRR 21-1229	11/3/21
B 3.8.3-3	0	Amend. No. 123	12/18/99
B 3.8.3-4	1	DRR 99-1624	12/18/99
B 3.8.3-5	0	Amend. No. 123	12/18/99
B 3.8.3-6	89	DRR 21-0966	7/7/21
B 3.8.3-7	12	DRR 02-1062	9/26/02
B 3.8.3-8	89	DRR 21-0966	7/7/21
B 3.8.3-9	0	Amend. No. 123	12/18/99
B 3.8.4-1	0	Amend. No. 123	12/18/99
B 3.8.4-2	0	Amend. No. 123	12/18/99
B 3.8.4-3	93	DRR 22-1363	2/2/23
B 3.8.4-4	0	Amend. No. 123	12/18/99
B 3.8.4-5	93	DRR 22-1363	2/2/23
B 3.8.4-6	89	DRR 21-0966	7/7/21
B 3.8.4-7	6	DRR 00-1541	3/13/01
B 3.8.4-8	93	DRR 22-1363	2/2/23
B 3.8.4-9	89	DRR 21-0966	7/7/21
B 3.8.5-1	57	DRR 13-0006	1/16/13
B 3.8.5-2	95	N/A	4/25/24
B 3.8.5-3	57	DRR 13-0006	1/16/13
B 3.8.5-4	57	DRR 13-0006	1/16/13
B 3.8.5-5	57	DRR 13-0006	1/16/13
B 3.8.6-1	0	Amend. No. 123	12/18/99
B 3.8.6-2	0	Amend. No. 123	12/18/99
B 3.8.6-3	89	DRR 21-0966	7/7/21
B 3.8.6-4	89	DRR 21-0966	7/7/21

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – B 3.8 ELECTRICAL POWER SYSTEMS (continued)			
B 3.8.6-5	0	Amend. No. 123	12/18/99
B 3.8.6-6	0	Amend. No. 123	12/18/99
B 3.8.7-1	69	DRR 15-0493	3/26/15
B 3.8.7-2	69	DRR 15-0493	3/26/15
B 3.8.7-3	89	DRR 21-0966	7/7/21
B 3.8.7-4	0	Amend. No. 123	12/18/99
B 3.8.8-1	57	DRR 13-0006	1/16/13
B 3.8.8-2	0	Amend. No. 123	12/18/99
B 3.8.8-3	69	DRR 15-0493	3/26/15
B 3.8.8-4	57	DRR 13-0006	1/16/13
B 3.8.8-5	93	DRR 22-1363	2/2/23
B 3.8.9-1	54	DRR 11-2394	11/16/11
B 3.8.9-2	69	DRR 15-0493	3/26/15
B 3.8.9-3	54	DRR 11-2394	11/16/11
B 3.8.9-4	0	Amend. No. 123	12/18/99
B 3.8.9-5	69	DRR 15-0493	3/26/15
B 3.8.9-6	0	Amend. No. 123	12/18/99
B 3.8.9-7	0	Amend. No. 123	12/18/99
B 3.8.9-8	89	DRR 21-0966	7/7/21
B 3.8.9-9	0	Amend. No. 123	12/18/99
B 3.8.10-1	57	DRR 13-0006	1/16/13
B 3.8.10-2	0	Amend. No. 123	12/18/99
B 3.8.10-3	0	Amend. No. 123	12/18/99
B 3.8.10-4	57	DRR 13-0006	1/16/13
B 3.8.10-5	57	DRR 13-0006	1/16/13
B 3.8.10-6	89	DRR 21-0966	7/7/21
TAB – B 3.9 REFUELING OPERATIONS			
B 3.9.1-1	0	Amend. No. 123	12/18/99
B 3.9.1-2	19	DRR 04-1414	10/12/04
B 3.9.1-3	19	DRR 04-1414	10/12/04
B 3.9.1-4	89	DRR 21-0966	7/7/21
B 3.9.2-1	0	Amend. No. 123	12/18/99
B 3.9.2-2	0	Amend. No. 123	12/18/99
B 3.9.2-3	89	DRR 21-0966	7/7/21
B 3.9.3-1	68	DRR 15-0248	2/26/15
B 3.9.3-2	68	DRR 15-0248	2/26/15
B 3.9.3-3	89	DRR 21-0966	7/7/21
B 3.9.3-4	89	DRR 21-0966	7/7/21
B 3.9.4-1	81	DRR 19-1027	10/28/19
B 3.9.4-2	13	DRR 02-1458	12/03/02
B 3.9.4-3	81	DRR 19-1027	10/28/19
B 3.9.4-4	23	DRR 05-1995	9/28/05
B 3.9.4-5	89	DRR 21-0966	7/7/21
B 3.9.4-6	89	DRR 21-0966	7/7/21
B 3.9.5-1	0	Amend. No. 123	12/18/99
B 3.9.5-2	72	DRR 15-1918	10/26/15
B 3.9.5-3	32	DRR 07-0139	2/7/07
B 3.9.5-4	89	DRR 21-0966	7/7/21
B 3.9.5-5	89	DRR 21-0966	7/7/21
B 3.9.6-1	0	Amend. No. 123	12/18/99

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – B 3.9 REFUELING OPERATIONS (continued)			
B 3.9.6-2	90	DRR 21-1229	11/3/21
B 3.9.6-3	42	DRR 09-1009	7/16/09
B 3.9.6-4	89	DRR 21-0966	7/7/21
B 3.9.6-5	89	DRR 21-0966	7/7/21
B 3.9.7-1	81	DRR 19-1027	10/28/19
B 3.9.7-2	89	DRR 21-0966	7/7/21
B 3.9.7-3	81	DRR 19-1027	10/28/19

Note 1 The page number is listed on the center of the bottom of each page.

Note 2 The revision number is listed in the lower right hand corner of each page. The Revision number will be page specific.

Note 3 The change document will be the document requesting the change. Amendment No. 123 issued the improved Technical Specifications and associated Bases which affected each page. The NRC has indicated that Bases changes will not be issued with License Amendments. As of Revision 94, the TS Bases is no longer controlled in PMAC. Therefore, starting with Revision 94, the DRR number will be N/A.

Note 4 The date effective or implemented is the date the Bases pages are issued by Document Control.

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SURVEILLANCE REQUIREMENTS

SR 3.3.1.16 (continued)

channels with response times listed as N.A. No response time testing requirements apply where N.A. is listed in Table B 3.3.1-2. Individual component response times are not modeled in the analyses. The analyses model the overall or total elapsed time, from the point at which the parameter exceeds the trip setpoint value at the sensor until loss of stationary gripper coil voltage.

For channels that include dynamic transfer Functions (e.g., lag, lead/lag, rate/lag, etc.), the response time verification is performed with the time constants set at their nominal values. The response time may be measured by a series of overlapping tests, or other verification (e.g., Ref. 7), such that the entire response time is measured.

Response time may be verified by actual response time tests in any series of sequential, overlapping or total channel measurements, or by the summation of allocated response times with actual response time tests on the remainder of the channel. Allocations for response times may be obtained from: (1) historical records based on acceptable response time tests (hydraulic, noise, or power interrupt tests), (2) inplace, onsite, or offsite (e.g. vendor) test measurements, or (3) utilizing vendor engineering specifications. WCAP-13632-P-A Revision 2, “Elimination of Pressure Sensor Response Time Testing Requirements” (Ref. 7), provides the basis and methodology for using allocated sensor response times in the overall verification of the channel response time for specific sensors identified in the WCAP. Response time verification for other sensor types must be demonstrated by test.

WCAP-14036-P-A, Revision 1, “Elimination of Periodic Protection Channel Response Time Tests,” (Ref. 14) provides the basis and methodology for using allocated signal processing and actuation logic response times in the overall verification of the protection system channel response time. The allocations for sensor, signal conditioning, and actuation logic response times must be verified prior to placing the component in operational service and re-verified following maintenance that may adversely affect response time. In general, electrical repair work does not impact response time provided the parts used for repair are of the same type and value. Specific components identified in the WCAP may be replaced without verification testing. One example where response time could be affected is replacing the sensing assembly of a transmitter.

The response time may be verified for components that replace the components that were previously evaluated in Ref. 6 and Ref. 14, provided that the components have been evaluated in accordance with the NRC approved methodology as discussed in Attachment 1 to TSTF-569, “Methodology to Eliminate Pressure Sensor and Protection

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SURVEILLANCE REQUIREMENTS

SR 3.3.1.16 (continued)

Channel (for Westinghouse Plants only) Response Time Testing," (Ref. 15).

The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

SR 3.3.1.16 is modified by a Note stating that neutron detectors are excluded from RTS RESPONSE TIME testing. This Note is necessary because of the difficulty in generating an appropriate detector input signal. Excluding the detectors is acceptable because the principles of detector operation ensure a virtually instantaneous response. Response time of the neutron flux signal portion of the channel shall be measured from detector output or input to the first electronic component in the channel.

REFERENCES

1. USAR, Chapter 7.
2. USAR, Chapter 15.
3. IEEE-279-1971.
4. 10 CFR 50.49.
5. WCNOC Nuclear Safety Analysis Setpoint Methodology for the Reactor Protection System, (TR-89-0001).
6. WCAP-13632-P-A, Revision 2, "Elimination of Pressure Sensor Response Time Testing Requirements," January 1996.
7. WCAP-9226, "Reactor Core Response to Excessive Secondary Steam Releases," Revision 1, January 1978.
8. IE Information Notice 79-22, "Qualification of Control Systems," September 14, 1979.
9. "Wolf Creek Setpoint Methodology Report," SNP(KG)-492, August 29, 1984.
10. USAR, Table 15.0-4.
11. WCAP-14333-P-A, Revision 1, "Probabilistic Risk Analysis of the RPS and ESFAS Test Times and Completion Times," October 1998.

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REFERENCES (continued)

12. WCAP-15376-P-A, Revision 1, “Risk-Informed Assessment of the RTS and ESFAS Surveillance Test Intervals and Reactor Trip Breaker Test and Completion Times,” March 2003.

13. WOG-06-17, “WCAP-10271-P-A Justification for Bypass Test Time and Completion Time Technical Specification Changes for Reactor Trip on Turbine Trip (ITSWG Action item #314),” January 20, 2006.

14. WCAP-14036-P-A, Revision 1, “Elimination of Periodic Protection Channel Response Time Tests,” December 1995.

15. Attachment 1 to TSTF-569, “Methodology to Eliminate Pressure Sensor and Protection Channel (for Westinghouse Plants only) Response Time Testing.”

TABLE B 3.3.1-1
(Page 1 of 2)

FUNCTION	TRIP SETPOINT[a]
1. Manual Reactor Trip	NA
2. Power Range Neutron Flux	
a. High	≤ 109% of RTP
b. Low	≤ 25% of RTP
3. Power Range Neutron Flux High Positive Rate	≤ 4% of RTP with a time constant ≥ 2 seconds
4. Intermediate Range Neutron Flux	≤ 25% of RTP
5. Source Range Neutron Flux	≤ 10^5 cps
6. Overtemperature ΔT	See Table 3.3.1-1 Note 1
7. Overpower ΔT	See Table 3.3.1-1 Note 2
8. Pressurizer Pressure	
a. Low	≥ 1940 psig
b. High	≤ 2385 psig
9. Pressurizer Water level - High	≤ 92% of instrument span
10. Reactor Coolant Flow - Low	≥ 89.9% of Normalized Flow
11. Not Used	
12. Undervoltage RCPs	≥ 10578 Vac
13. Underfrequency RCPs	≥ 57.15 Hz
14. Steam Generator (SG) Water Level Low - Low	≥ 23.5% of narrow range instrument span
15. Not Used	
16. Turbine Trip	
a. Low Fluid Oil Pressure	≥ 590.00 psig
b. Turbine Stop Valve Closure	≥ 1% open

TABLE B 3.3.1-1
(Page 2 of 2)

FUNCTION	TRIP SETPOINT[a]
17. Safety Injection (SI) Input from Engineered Safety Feature Actuation System (ESFAS)	N.A.
18. Reactor Trip System Interlocks	
a. Intermediate Range Neutron Flux, P-6	≥ 1.0E-10 amps
b. Low Power Reactor Trips Block, P-7	N.A.
c. Power Range Neutron Flux, P-8	≤ 48% RTP
d. Power Range Neutron Flux, P-9	≤ 50% RTP
e. Power Range Neutron Flux, P-10	10% RTP
f. Turbine Impulse Pressure, P-13	≤ 10% turbine power
19. Reactor Trip Breakers	N.A.
20. Reactor Trip breaker Undervoltage and Shunt Trip Mechanisms	N.A.
21. Automatic Trip Logic	N.A.

[a] The inequality sign only indicates conservative direction. The as-left value will be within a two-sided calibration tolerance band on either side of the nominal value. This also applies to the Overtemperature ΔT and Overpower ΔT K and τ values.

TABLE B 3.3.1-2
(Page 1 of 2)

FUNCTIONAL UNIT	RESPONSE TIME
1. Manual Reactor Trip	N.A.
2. Power Range Neutron Flux	
a. High	≤ 0.5 second[1]
b. Low	≤ 0.5 second[1]
3. Power Range Neutron Flux High Positive Rate	≤ 0.5 second[1].
4. Intermediate Range Neutron Flux	N.A.
5. Source Range Neutron Flux	N.A.
6. Overtemperature ΔT	≤ 6.0 seconds[1]
7. Overpower ΔT	≤ 6.0 seconds[1]
8. Pressurizer Pressure	
a. Low	≤ 2.0 seconds
b. High	≤ 1.0 second
9. Pressurizer Water Level - High	N.A.
10. Reactor Coolant Flow - Low	
a. Single Loop (Above P-8)	≤ 1.0 second
b. Two Loops (Above P-7 and below P-8)	≤ 1.0 second
11. Not Used	
12. Undervoltage - Reactor Coolant Pumps	≤ 1.5 seconds
13. Underfrequency - Reactor Coolant Pumps	≤ 0.6 second
14. Steam Generator Water Level - Low-Low	≤ 2.0 seconds
15. Not Used	

[1] Response time of the neutron flux signal portion of the channel shall be measured from detector output or input of first electronic component in channel.

TABLE B 3.3.1-2
(Page 2 of 2)

FUNCTIONAL UNIT	RESPONSE TIME
16. Turbine Trip	
a. Low Fluid Oil Pressure	N.A.
b. Turbine Stop Valve Closure	N.A.
17. Safety Injection Input for ESF	N.A.
18. Reactor Trip System Interlocks	N.A.
19. Reactor Trip Breakers	N.A.
20. Reactor Trip Breaker Undervoltage and Shunt Trip Mechanisms	N.A.
21. Automatic Trip and Interlock Logic	N.A.

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SURVEILLANCE REQUIREMENTS

SR 3.3.2.10 (continued)

response times in the overall verification of the channel response time for specific sensors identified in the WCAP. Response time verification for other sensor types must be demonstrated by test.

WCAP-14036-P-A, Revision 1, "Elimination of Periodic Protection Channel Response Time Tests," (Ref. 15) provides the basis and methodology for using allocated signal processing and actuation logic response times in the overall verification of the protection system channel response time. The allocations for sensor, signal conditioning, and actuation logic response times must be verified prior to placing the component in operational service and re-verified following maintenance that may adversely affect response time. In general, electrical repair work does not impact response time provided the parts used for repair are of the same type and value. Specific components identified in the WCAP may be replaced without verification testing. One example where response time could be affected is replacing the sensing assembly of a transmitter.

The response time may be verified for components that replace the components that were previously evaluated in Ref. 8 and Ref. 15, provided that the components have been evaluated in accordance with the NRC approved methodology as discussed in Attachment 1 to TSTF-569, "Methodology to Eliminate Pressure Sensor and Protection Channel (for Westinghouse Plants only) Response Time Testing," (Ref. 16).

The NRC approved the use of ASME Code Case OMN-1, "Alternative Rules for Preservice and Inservice Testing of Certain Electric Motor-Operated Valve Assemblies in Light-Water Reactor Plants," as an alternative to stroke time testing for motor-operated valves (Ref. 14). The parameters that must be present to achieve the analyzed response time under design basis conditions are measured to ensure the valve is capable of performing its safety function. This process verifies design basis capability, including response time, and is a significant improvement over simple stroke time measurement. This process allows the establishment of periodic valve test intervals if there is assurance that the valve will remain capable of performing its safety function throughout the interval.

ESF response times specified in Table B 3.3.2-2 which include sequential operation of RWST and VCT valves (Notes 3 and 4) are based on values assumed in the non-LOCA safety analyses. These analyses take credit for injection of borated water from the RWST. Injection of borated water is assumed not to occur until the VCT charging pump suction valves are closed following opening of the RWST charging pump suction valves.

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SURVEILLANCE REQUIREMENTS

SR 3.3.2.10 (continued)

When the sequential operation of the RWST and VCT valves is not included in the response times (Note 7), the values specified are based on the LOCA analyses. The LOCA analyses take credit for injection flow regardless of the source. Verification of the response times specified in Table B 3.3.2-2 will assure that the assumptions used for the LOCA and non-LOCA analyses with respect to the operation of the VCT and RWST valves are valid.

The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

This SR is modified by a Note that clarifies that the turbine driven AFW pump is tested within 24 hours after reaching 900 psig in the SGs.

SR 3.3.2.11

3.3.2.11 is the performance of a TADOT as described in SR 3.3.2.8, except that it is performed for the P-4 Reactor Trip Interlock. The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

The SR is modified by a Note that excludes verification of setpoints during the TADOT. The Function tested has no associated setpoint. This TADOT does not include the circuitry associated with steam dump operation since it is control grade circuitry.

SR 3.3.2.12

SR 3.3.2.12 is the performance of a monthly COT on ESFAS Function 6.h, “Auxiliary Feedwater Pump Suction Transfer on Suction Pressure - Low.”

A COT is performed to ensure the channel will perform the intended Function. Setpoints must be found within the Allowable Values specified in Table 3.3.2-1.

The setpoint shall be left set consistent with the assumptions of the current unit specific setpoint methodology.

The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

BASES

REFERENCES

1. USAR, Chapter 6.

2. USAR, Chapter 7.

3. USAR, Chapter 15.

4. IEEE-279-1971.

5. 10 CFR 50.49.

6. WCNOC Nuclear Safety Analysis Setpoint Methodology for the Reactor Protection System, TR-89-0001.

7. WCAP-10271-P-A Supplement 2, Revision 1, "Evaluation of Surveillance Frequencies and Out of Service Times for the Engineered Safety Features Actuation System," June 1990.

8. WCAP-13632-P-A, Revision 2, “Elimination of Pressure Sensor Response Time Testing Requirements,” January 1996.

9. “Wolf Creek Setpoint Methodology Report,” SNP (KG)-492, August 29, 1984.

10. Amendment No. 43 to Facility Operating License No. NPF-42, March 29, 1991.

11. WCAP-14333-P-A, Revision 1, “Probabilistic Risk Analysis of the RPS and ESFAS Test Times and Completion Times,” October 1998.

12. 10 CFR 50.55a(b)(3)(iii), Code Case OMN-1.

13. Performance Improvement Request (PIR) 2005-2067.

14. Amendment No. 231 to Renewed Facility Operating License No. NPF-42, February 23, 2022.

15. WCAP-14036-P-A, Revision 1, “Elimination of Periodic Protection Channel Response Time Tests,” December 1995.

16. Attachment 1 to TSTF-569, “Methodology to Eliminate Pressure Sensor and Protection Channel (for Westinghouse Plants only) Response Time Testing.”

TECHNICAL SPECIFICATION BASES

Wolf Creek Generating Station, Unit 1

Summary of Revision 98:

1) Revised TS Bases Section B 3.2.1 as part of incorporation of License Amendment 244. Amendment 244 revises TS 3.2.1, "Heat Flux Hot Channel Factor ($F_Q(Z)$) (F_Q Methodology)," to implement the methodology in Pressurized Water Reactor Owners Group Topical Report WCAP-17661-P-A, Revision 1, "Improved RAOC [Relaxed Axial Offset Control] and CAOC [Constant Axial Offset Control] F_Q Surveillance Technical Specifications."

2) Revised TS Bases B 3.4.15 to remove the requirement that the plant process computer must be capable of calculating the leakage rate indicated by increases in sump level. This will allow for manually calculating RCS leak rate once per shift when the plant process computer is out of service. This is consistent with the Wolf Creek Technical Specifications (TS) prior to the adoption of the Improved TS (ITS), when Surveillance Requirement 4.4.6.2.1.b was deleted. Because this was an approved method for meeting the LCO (current ITS LCO is 3.4.15) prior to adoption of ITS, this method may still be used to meet the LCO,

3) Revised TS Bases page B 3.8.1-1 to indicate that with the addition of the Blackberry line, offsite power is now supplied to the unit switchyard from the transmission network by four transmission lines.

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – Title Page Technical Specification			
Cover Page			
Title Page			
TAB – Table of Contents			
i	98	N/A	10/15/25
ii	81	DRR 19-1027	10/28/19
iii	81	DRR 19-1027	10/28/19
TAB – B 2.0 SAFETY LIMITS (SLs)			
B 2.1.1-1	0	Amend. No. 123	12/18/99
B 2.1.1-2	14	DRR 03-0102	2/12/03
B 2.1.1-3	14	DRR 03-0102	2/12/03
B 2.1.1-4	14	DRR 03-0102	2/12/03
B 2.1.2-1	84	DRR 20-0400	08/18/20
B 2.1.2-2	84	DRR 20-0400	08/18/20
B 2.1.2-3	81	DRR 19-1027	10/28/19
TAB – B 3.0 LIMITING CONDITION FOR OPERATION (LCO) APPLICABILTY			
B 3.0-1	81	DRR 19-1027	10/28/19
B 3.0-2	0	Amend. No. 123	12/18/99
B 3.0-3	81	DRR 19-1027	10/28/19
B 3.0-4	81	DRR 19-1027	10/28/19
B 3.0-5	81	DRR 19-1027	10/28/19
B 3.0-6	81	DRR 19-1027	10/28/19
B 3.0-7	81	DRR 19-1027	10/28/19
B 3.0-8	81	DRR 19-1027	10/28/19
B 3.0-9	81	DRR 19-1027	10/28/19
B 3.0-10	81	DRR 19-1027	10/28/19
B 3.0-11	81	DRR 19-1027	10/28/19
B 3.0-12	81	DRR 19-1027	10/28/19
B 3.0-13	81	DRR 19-1027	10/28/19
B 3.0-14	81	DRR 19-1027	10/28/19
B 3.0-15	81	DRR 19-1027	10/28/19
B 3.0-16	81	DRR 19-1027	10/28/19
B 3.0-17	81	DRR 19-1027	10/28/19
TAB – B 3.1 REACTIVITY CONTROL SYSTEMS			
B 3.1.1-1	0	Amend. No. 123	12/18/99
B 3.1.1-2	0	Amend. No. 123	12/18/99
B 3.1.1-3	95	N/A	4/25/24
B 3.1.1-4	81	DRR 19-1027	10/28/19
B 3.1.1-5	89	DRR 21-0966	7/7/21
B 3.1.2-1	0	Amend. No. 123	12/18/99
B 3.1.2-2	0	Amend. No. 123	12/18/99
B 3.1.2-3	0	Amend. No. 123	12/18/99
B 3.1.2-4	0	Amend. No. 123	12/18/99
B 3.1.2-5	89	DRR 21-0966	7/7/21
B 3.1.3-1	96	N/A	12/11/2024
B 3.1.3-2	96	N/A	12/11/2024
B 3.1.3-3	96	N/A	12/11/2024

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – B 3.1 REACTIVITY CONTROL SYSTEMS (continued)			
B 3.1.3-4	96	N/A	12/11/2024
B 3.1.3-5	96	N/A	12/11/2024
B 3.1.3-6	95	N/A	4/25/24
B 3.1.4-1	0	Amend. No. 123	12/18/99
B 3.1.4-2	0	Amend. No. 123	12/18/99
B 3.1.4-3	48	DRR 10-3740	12/28/10
B 3.1.4-4	0	Amend. No. 123	12/18/99
B 3.1.4-5	0	Amend. No. 123	12/18/99
B 3.1.4-6	48	DRR 10-3740	12/28/10
B 3.1.4-7	0	Amend. No. 123	12/18/99
B 3.1.4-8	89	DRR 21-0966	7/7/21
B 3.1.4-9	89	DRR 21-0966	7/7/21
B 3.1.5-1	0	Amend. No. 123	12/18/99
B 3.1.5-2	0	Amend. No. 123	12/18/99
B 3.1.5-3	0	Amend. No. 123	12/18/99
B 3.1.5-4	89	DRR 21-0966	7/7/21
B 3.1.6-1	0	Amend. No. 123	12/18/99
B 3.1.6-2	0	Amend. No. 123	12/18/99
B 3.1.6-3	0	Amend. No. 123	12/18/99
B 3.1.6-4	0	Amend. No. 123	12/18/99
B 3.1.6-5	89	DRR 21-0966	7/7/21
B 3.1.6-6	95	N/A	4/25/24
B 3.1.7-1	0	Amend. No. 123	12/18/99
B 3.1.7-2	0	Amend. No. 123	12/18/99
B 3.1.7-3	48	DRR 10-3740	12/28/10
B 3.1.7-4	48	DRR 10-3740	12/28/10
B 3.1.7-5	48	DRR 10-3740	12/28/10
B 3.1.7-6	0	Amend. No. 123	12/18/99
B 3.1.8-1	0	Amend. No. 123	12/18/99
B 3.1.8-2	0	Amend. No. 123	12/18/99
B 3.1.8-3	15	DRR 03-0860	7/10/03
B 3.1.8-4	15	DRR 03-0860	7/10/03
B 3.1.8-5	89	DRR 21-0966	7/7/21
B 3.1.8-6	89	DRR 21-0966	7/7/21
B 3.1.9-1	84	DRR 20-0400	08/18/20
B 3.1.9-2	84	DRR 20-0400	08/18/20
B 3.1.9-3	84	DRR 20-0400	08/18/20
B 3.1.9-4	84	DRR 20-0400	08/18/20
B 3.1.9-5	89	DRR 21-0966	7/7/21
TAB – B 3.2 POWER DISTRIBUTION LIMITS			
B 3.2.1-1	98	N/A	10/15/25
B 3.2.1-2	98	N/A	10/15/25
B 3.2.1-3	98	N/A	10/15/25
B 3.2.1-4	98	N/A	10/15/25
B 3.2.1-5	98	N/A	10/15/25
B 3.2.1-6	98	N/A	10/15/25
B 3.2.1-7	98	N/A	10/15/25
B 3.2.1-8	98	N/A	10/15/25
B 3.2.1-9	98	N/A	10/15/25
B 3.2.1-10	98	N/A	10/15/25

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – B 3.2 POWER DISTRIBUTION LIMITS (continued)			
B 3.2.1-11 (new)	98	N/A	10/15/25
B 3.2.1-12 (new)	98	N/A	10/15/25
B 3.2.1-13 (new)	98	N/A	10/15/25
B 3.2.1-14 (new)	98	N/A	10/15/25
B 3.2.2-1	48	DRR 10-3740	12/28/10
B 3.2.2-2	0	Amend. No. 123	12/18/99
B 3.2.2-3	48	DRR 10-3740	12/28/10
B 3.2.2-4	48	DRR 10-3740	12/28/10
B 3.2.2-5	48	DRR 10-3740	12/28/10
B 3.2.2-6	89	DRR 21-0966	7/7/21
B 3.2.3-1	0	Amend. No. 123	12/18/99
B 3.2.3-2	0	Amend. No. 123	12/18/99
B 3.2.3-3	89	DRR 21-0966	7/7/21
B 3.2.4-1	0	Amend. No. 123	12/18/99
B 3.2.4-2	0	Amend. No. 123	12/18/99
B 3.2.4-3	48	DRR 10-3740	12/28/10
B 3.2.4-4	0	Amend. No. 123	12/18/99
B 3.2.4-5	48	DRR 10-3740	12/28/10
B 3.2.4-6	89	DRR 21-0966	7/7/21
B 3.2.4-7	89	DRR 21-0966	7/7/21
TAB – B 3.3 INSTRUMENTATION			
B 3.3.1-1	84	DRR 20-0400	08/18/20
B 3.3.1-2	0	Amend. No. 123	12/18/99
B 3.3.1-3	0	Amend. No. 123	12/18/99
B 3.3.1-4	0	Amend. No. 123	12/18/99
B 3.3.1-5	0	Amend. No. 123	12/18/99
B 3.3.1-6	0	Amend. No. 123	12/18/99
B 3.3.1-7	5	DRR 00-1427	10/12/00
B 3.3.1-8	0	Amend. No. 123	12/18/99
B 3.3.1-9	84	DRR 20-0400	08/18/20
B 3.3.1-10	84	DRR 20-0400	08/18/20
B 3.3.1-11	95	N/A	4/25/24
B 3.3.1-12	95	N/A	4/25/24
B 3.3.1-13	95	N/A	4/25/24
B 3.3.1-14	95	N/A	4/25/24
B 3.3.1-15	95	N/A	4/25/24
B 3.3.1-16	95	N/A	4/25/24
B 3.3.1-17	95	N/A	4/25/24
B 3.3.1-18	95	N/A	4/25/24
B 3.3.1-19	95	N/A	4/25/24
B 3.3.1-20	84	DRR 20-0400	08/18/20
B 3.3.1-21	84	DRR 20-0400	08/18/20
B 3.3.1-22	84	DRR 20-0400	08/18/20
B 3.3.1-23	84	DRR 20-0400	08/18/20
B 3.3.1-24	84	DRR 20-0400	08/18/20
B 3.3.1-25	84	DRR 20-0400	08/18/20
B 3.3.1-26	84	DRR 20-0400	08/18/20
B 3.3.1-27	84	DRR 20-0400	08/18/20
B 3.3.1-28	84	DRR 20-0400	08/18/20
B 3.3.1-29	84	DRR 20-0400	08/18/20

PAGE (1)	REVISION NO. (2)	CHANGE DOCUMENT (3)	DATE EFFECTIVE/ IMPLEMENTED (4)
TAB – B 3.3 INSTRUMENTATION (continued)			
B 3.3.1-30	84	DRR 20-0400	08/18/20
B 3.3.1-31	84	DRR 20-0400	08/18/20
B 3.3.1-32	84	DRR 20-0400	08/18/20
B 3.3.1-33	84	DRR 20-0400	08/18/20
B 3.3.1-34	84	DRR 20-0400	08/18/20
B 3.3.1-35	84	DRR 20-0400	08/18/20
B 3.3.1-36	95	N/A	4/25/24
B 3.3.1-37	84	DRR 20-0400	08/18/20
B 3.3.1-38	84	DRR 20-0400	08/18/20
B 3.3.1-39	84	DRR 20-0400	08/18/20
B 3.3.1-40	84	DRR 20-0400	08/18/20
B 3.3.1-41	84	DRR 20-0400	08/18/20
B 3.3.1-42	84	DRR 20-0400	08/18/20
B 3.3.1-43	84	DRR 20-0400	08/18/20
B 3.3.1-44	84	DRR 20-0400	08/18/20
B 3.3.1-45	84	DRR 20-0400	08/18/20
B 3.3.1-46	84	DRR 20-0400	08/18/20
B 3.3.1-47	89	DRR 21-0966	7/7/21
B 3.3.1-48	84	DRR 20-0400	08/18/20
B 3.3.1-49	89	DRR 21-0966	7/7/21
B 3.3.1-50	89	DRR 21-0966	7/7/21
B 3.3.1-51	89	DRR 21-0966	7/7/21
B 3.3.1-52	89	DRR 21-0966	7/7/21
B 3.3.1-53	89	DRR 21-0966	7/7/21
B 3.3.1-54	89	DRR 21-0966	7/7/21
B 3.3.1-55	89	DRR 21-0966	7/7/21
B 3.3.1-56	97	N/A	5/27/25
B 3.3.1-57	97	N/A	5/27/25
B 3.3.1-58	97	N/A	5/27/25
B 3.3.1-59	97	N/A	5/27/25
B 3.3.1-60	97	N/A	5/27/25
B 3.3.1-61	97	N/A	5/27/25
B 3.3.1-62	97	N/A	5/27/25
B 3.3.2-1	84	DRR 20-0400	08/18/20
B 3.3.2-2	0	Amend. No. 123	12/18/99
B 3.3.2-3	0	Amend. No. 123	12/18/99
B 3.3.2-4	0	Amend. No. 123	12/18/99
B 3.3.2-5	0	Amend. No. 123	12/18/99
B 3.3.2-6	7	DRR 01-0474	5/1/01
B 3.3.2-7	0	Amend. No. 123	12/18/99
B 3.3.2-8	0	Amend. No. 123	12/18/99
B 3.3.2-9	0	Amend. No. 123	12/18/99
B 3.3.2-10	0	Amend. No. 123	12/18/99
B 3.3.2-11	0	Amend. No. 123	12/18/99
B 3.3.2-12	81	DRR 19-1027	10/28/19
B 3.3.2-13	0	Amend. No. 123	12/18/99
B 3.3.2-14	2	DRR 00-0147	4/24/00
B 3.3.2-15	0	Amend. No. 123	12/18/99
B 3.3.2-16	0	Amend. No. 123	12/18/99
B 3.3.2-17	0	Amend. No. 123	12/18/99
B 3.3.2-18	0	Amend. No. 123	12/18/99

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TAB – B 3.3 INSTRUMENTATION (continued)			
B 3.3.2-19	37	DRR 08-0503	4/8/08
B 3.3.2-20	37	DRR 08-0503	4/8/08
B 3.3.2-21	37	DRR 08-0503	4/8/08
B 3.3.2-23	37	DRR 08-0503	4/8/08
B 3.3.2-24	39	DRR 08-1096	8/28/08
B 3.3.2-25	39	DRR 08-1096	8/28/08
B 3.3.2-26	39	DRR 08-1096	8/28/08
B 3.3.2-27	37	DRR 08-0503	4/8/08
B 3.3.2-28	84	DRR 20-0400	08/18/20
B 3.3.2-29	0	Amend. No. 123	12/18/99
B 3.3.2-30	0	Amend. No. 123	12/18/99
B 3.3.2-31	52	DRR 11-0724	4/11/11
B 3.3.2-32	52	DRR 11-0724	4/11/11
B 3.3.2-33	0	Amend. No. 123	12/18/99
B 3.3.2-34	0	Amend. No. 123	12/18/99
B 3.3.2-35	20	DRR 04-1533	2/16/05
B 3.3.2-36	20	DRR 04-1533	2/16/05
B 3.3.2-37	20	DRR 04-1533	2/16/05
B 3.3.2-38	20	DRR 04-1533	2/16/05
B 3.3.2-39	25	DRR 06-0800	5/18/06
B 3.3.2-40	20	DRR 04-1533	2/16/05
B 3.3.2-41	45	Amend. No. 187 (ETS)	3/5/10
B 3.3.2-42	45	Amend. No. 187 (ETS)	3/5/10
B 3.3.2-43	20	DRR 04-1533	2/16/05
B 3.3.2-44	91	DRR 22-0243	3/2/22
B 3.3.2-45	92	DRR 22-0767	11/3/22
B 3.3.2-46	89	DRR 21-0966	7/7/21
B 3.3.2-47	89	DRR 21-0966	7/7/21
B 3.3.2-48	89	DRR 21-0966	7/7/21
B 3.3.2-49	89	DRR 21-0966	7/7/21
B 3.3.2-50	97	N/A	5/27/25
B 3.3.2-51	97	N/A	5/27/25
B 3.3.2-52	97	N/A	5/27/25
B 3.3.2-53	43	DRR 09-1416	9/2/09
B 3.3.2-54	43	DRR 09-1416	9/2/09
B 3.3.2-55	43	DRR 09-1416	9/2/09
B 3.3.2-56	43	DRR 09-1416	9/2/09
B 3.3.2-57	43	DRR 09-1416	9/2/09
B 3.3.3-1	0	Amend. No. 123	12/18/99
B 3.3.3-2	5	DRR 00-1427	10/12/00
B 3.3.3-3	0	Amend. No. 123	12/18/99
B 3.3.3-4	0	Amend. No. 123	12/18/99
B 3.3.3-5	0	Amend. No. 123	12/18/99
B 3.3.3-6	8	DRR 01-1235	9/19/01
B 3.3.3-7	21	DRR 05-0707	4/20/05
B 3.3.3-8	81	DRR 19-1027	10/28/19
B 3.3.3-9	8	DRR 01-1235	9/19/01
B 3.3.3-10	19	DRR 04-1414	10/12/04
B 3.3.3-11	19	DRR 04-1414	10/12/04
B 3.3.3-12	21	DRR 05-0707	4/20/05
B 3.3.3-13	89	DRR 21-0966	7/7/21

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TAB – B 3.3 INSTRUMENTATION (continued)			
B 3.3.3-14	89	DRR 21-0966	7/7/21
B 3.3.3-15	8	DRR 01-1235	9/19/01
B 3.3.4-1	0	Amend. No. 123	12/18/99
B 3.3.4-2	9	DRR 02-1023	2/28/02
B 3.3.4-3	15	DRR 03-0860	7/10/03
B 3.3.4-4	19	DRR 04-1414	10/12/04
B 3.3.4-5	89	DRR 21-0966	7/7/21
B 3.3.4-6	89	DRR 21-0966	7/7/21
B 3.3.5-1	88	DRR 21-0591	4/28/21
B 3.3.5-2	88	DRR 21-0591	4/28/21
B 3.3.5-3	1	DRR 99-1624	12/18/99
B 3.3.5-4	1	DRR 99-1624	12/18/99
B 3.3.5-5	0	Amend. No. 123	12/18/99
B 3.3.5-6	89	DRR 21-0966	7/7/21
B 3.3.5-7	89	DRR 21-0966	7/7/21
B 3.3.6-1	81	DRR 19-1027	10/28/19
B 3.3.6-2	81	DRR 19-1027	10/28/19
B 3.3.6-3	0	Amend. No. 123	12/18/99
B 3.3.6-4	0	Amend. No. 123	12/18/99
B 3.3.6-5	89	DRR 21-0966	7/7/21
B 3.3.6-6	89	DRR 21-0966	7/7/21
B 3.3.6-7	89	DRR 21-0966	7/7/21
B 3.3.7-1	81	DRR 19-1027	10/28/19
B 3.3.7-2	81	DRR 19-1027	10/28/19
B 3.3.7-3	57	DRR 13-0006	1/16/13
B 3.3.7-4	0	Amend. No. 123	12/18/99
B 3.3.7-5	0	Amend. No. 123	12/18/99
B 3.3.7-6	89	DRR 21-0966	7/7/21
B 3.3.7-7	89	DRR 21-0966	7/7/21
B 3.3.7-8	89	DRR 21-0966	7/7/21
B 3.3.8-1	84	DRR 20-0400	8/18/20
B 3.3.8-2	0	Amend. No. 123	12/18/99
B 3.3.8-3	57	DRR 13-0006	1/16/13
B 3.3.8-4	57	DRR 13-0006	1/16/13
B 3.3.8-5	89	DRR 21-0966	7/7/21
B 3.3.8-6	89	DRR 21-0966	7/7/21
B 3.3.8-7	89	DRR 21-0966	7/7/21
TAB – B 3.4 REACTOR COOLANT SYSTEM (RCS)			
B 3.4.1-1	84	DRR 20-0400	08/18/20
B 3.4.1-2	84	DRR 20-0400	08/18/20
B 3.4.1-3	10	DRR 02-0411	4/5/02
B 3.4.1-4	0	Amend. No. 123	12/18/99
B 3.4.1-5	89	DRR 21-0966	7/7/21
B 3.4.1-6	89	DRR 21-0966	7/7/21
B 3.4.2-1	0	Amend. No. 123	12/18/99
B 3.4.2-2	0	Amend. No. 123	12/18/99
B 3.4.2-3	89	DRR 21-0966	7/7/21
B 3.4.3-1	67	DRR 15-0116	2/10/15
B 3.4.3-2	0	Amend. No. 123	12/18/99
B 3.4.3-3	0	Amend. No. 123	12/18/99

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TAB – B 3.4 REACTOR COOLANT SYSTEM (RCS) (continued)			
B 3.4.3-4	0	Amend. No. 123	12/18/99
B 3.4.3-5	0	Amend. No. 123	12/18/99
B 3.4.3-6	89	DRR 21-0966	7/7/21
B 3.4.3-7	0	Amend. No. 123	12/18/99
B 3.4.4-1	0	Amend. No. 123	12/18/99
B 3.4.4-2	29	DRR 06-1984	10/17/06
B 3.4.4-3	89	DRR 21-0966	7/7/21
B 3.4.5-1	0	Amend. No. 123	12/18/99
B 3.4.5-2	53	DRR 11-1513	7/18/11
B 3.4.5-3	29	DRR 06-1984	10/17/06
B 3.4.5-4	0	Amend. No. 123	12/18/99
B 3.4.5-5	89	DRR 21-0966	7/7/21
B 3.4.5-6	89	DRR 21-0966	7/7/21
B 3.4.6-1	53	DRR 11-1513	7/18/11
B 3.4.6-2	72	DRR 15-1918	10/26/15
B 3.4.6-3	12	DRR 02-1062	9/26/02
B 3.4.6-4	89	DRR 21-0966	7/7/21
B 3.4.6-5	75	DRR 16-1909	10/26/16
B 3.4.6-6	89	DRR 21-0966	7/7/21
B 3.4.7-1	12	DRR 02-1062	9/26/02
B 3.4.7-2	17	DRR 04-0453	5/26/04
B 3.4.7-3	90	DRR 21-1229	11/3/21
B 3.4.7-4	89	DRR 21-0966	7/7/21
B 3.4.7-5	89	DRR 21-0966	7/7/21
B 3.4.7-6	89	DRR 21-0966	7/7/21
B 3.4.8-1	53	DRR 11-1513	7/18/11
B 3.4.8-2	90	DRR 21-1229	11/3/21
B 3.4.8-3	89	DRR 21-0966	7/7/21
B 3.4.8-4	89	DRR 21-0966	7/7/21
B 3.4.8-5	89	DRR 21-0966	7/7/21
B 3.4.9-1	0	Amend. No. 123	12/18/99
B 3.4.9-2	0	Amend. No. 123	12/18/99
B 3.4.9-3	0	Amend. No. 123	12/18/99
B 3.4.9-4	89	DRR 21-0966	7/7/21
B 3.4.10-1	5	DRR 00-1427	10/12/00
B 3.4.10-2	5	DRR 00-1427	10/12/00
B 3.4.10-3	0	Amend. No. 123	12/18/99
B 3.4.10-4	32	DRR 07-0139	2/7/07
B 3.4.11-1	0	Amend. No. 123	12/18/99
B 3.4.11-2	1	DRR 99-1624	12/18/99
B 3.4.11-3	19	DRR 04-1414	10/12/04
B 3.4.11-4	0	Amend. No. 123	12/18/99
B 3.4.11-5	1	DRR 99-1624	12/18/99
B 3.4.11-6	0	Amend. No. 123	12/18/99
B 3.4.11-7	89	DRR 21-0966	7/7/21
B 3.4.12-1	61	DRR 14-0346	2/27/14
B 3.4.12-2	61	DRR 14-0346	2/27/14
B 3.4.12-3	0	Amend. No. 123	12/18/99
B 3.4.12-4	61	DRR 14-0346	2/27/14
B 3.4.12-5	61	DRR 14-0346	2/27/14
B 3.4.12-6	56	DRR 12-1792	11/7/12

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TAB – B 3.4 REACTOR COOLANT SYSTEM (RCS) (continued)			
B 3.4.12-7	61	DRR 14-0346	2/27/14
B 3.4.12-8	1	DRR 99-1624	12/18/99
B 3.4.12-9	56	DRR 12-1792	11/7/12
B 3.4.12-10	0	Amend. No. 123	12/18/99
B 3.4.12-11	61	DRR 14-0346	2/27/14
B 3.4.12-12	89	DRR 21-0966	7/7/21
B 3.4.12-13	89	DRR 21-0966	7/7/21
B 3.4.12-14	89	DRR 21-0966	7/7/21
B 3.4.13-1	0	Amend. No. 123	12/18/99
B 3.4.13-2	94	N/A	10/6/23
B 3.4.13-3	94	N/A	10/6/23
B 3.4.13-4	94	N/A	10/6/23
B 3.4.13-5	94	N/A	10/6/23
B 3.4.13-6	89	DRR 21-0966	7/7/21
B 3.4.14-1	0	Amend. No. 123	12/18/99
B 3.4.14-2	0	Amend. No. 123	12/18/99
B 3.4.14-3	0	Amend. No. 123	12/18/99
B 3.4.14-4	0	Amend. No. 123	12/18/99
B 3.4.14-5	89	DRR 21-0966	7/7/21
B 3.4.14-6	89	DRR 21-0966	7/7/21
B 3.4.15-1	31	DRR 06-2494	12/13/06
B 3.4.15-2	31	DRR 06-2494	12/13/06
B 3.4.15-3	98	N/A	10/15/25
B 3.4.15-4	98	N/A	10/15/25
B 3.4.15-5	98	N/A	10/15/25
B 3.4.15-6	98	N/A	10/15/25
B 3.4.15-7	98	N/A	10/15/25
B 3.4.15-8	98	N/A	10/15/25
B 3.4.16-1	82	DRR 20-0077	1/29/20
B 3.4.16-2	84	DRR 20-0400	08/18/20
B 3.4.16-3	82	DRR 20-0077	1/29/20
B 3.4.16-4	92	DRR 22-0767	11/3/22
B 3.4.16-5	92	DRR 22-0767	11/3/22
B 3.4.17-1	29	DRR 06-1984	10/17/06
B 3.4.17-2	81	DRR 19-1027	10/28/19
B 3.4.17-3	29	DRR 06-1984	10/17/06
B 3.4.17-4	81	DRR 19-1027	10/28/19
B 3.4.17-5	57	DRR 13-0006	1/16/13
B 3.4.17-6	57	DRR 13-0006	1/16/13
B 3.4.17-7	81	DRR 19-1027	10/28/19
TAB – B 3.5 EMERGENCY CORE COOLING SYSTEMS (ECCS)			
B 3.5.1-1	0	Amend. No. 123	12/18/99
B 3.5.1-2	0	Amend. No. 123	12/18/99
B 3.5.1-3	73	DRR 15-2135	11/17/15
B 3.5.1-4	73	DRR 15-2135	11/17/15
B 3.5.1-5	1	DRR 99-1624	12/18/99
B 3.5.1-6	89	DRR 21-0966	7/7/21
B 3.5.1-7	89	DRR 21-0966	7/7/21
B 3.5.1-8	1	DRR 99-1624	12/18/99
B 3.5.2-1	84	DRR 20-0400	08/18/20

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TAB – B 3.5 EMERGENCY CORE COOLING SYSTEMS (ECCS) (continued)			
B 3.5.2-2	0	Amend. No. 123	12/18/99
B 3.5.2-3	0	Amend. No. 123	12/18/99
B 3.5.2-4	0	Amend. No. 123	12/18/99
B 3.5.2-5	72	DRR 15-1918	10/26/15
B 3.5.2-6	42	DRR 09-1009	7/16/09
B 3.5.2-7	89	DRR 21-0966	7/7/21
B 3.5.2-8	89	DRR 21-0966	7/7/21
B 3.5.2-9	89	DRR 21-0966	7/7/21
B 3.5.2-10	89	DRR 21-0966	7/7/21
B 3.5.2-11	89	DRR 21-0966	7/7/21
B 3.5.2-12	72	DRR 15-1918	10/26/15
B 3.5.3-1	56	DRR 12-1792	11/7/12
B 3.5.3-2	72	DRR 15-1918	10/26/15
B 3.5.3-3	56	DRR 12-1792	11/7/12
B 3.5.3-4	56	DRR 12-1792	11/7/12
B 3.5.4-1	0	Amend. No. 123	12/18/99
B 3.5.4-2	0	Amend. No. 123	12/18/99
B 3.5.4-3	0	Amend. No. 123	12/18/99
B 3.5.4-4	0	Amend. No. 123	12/18/99
B 3.5.4-5	89	DRR 21-0966	7/7/21
B 3.5.4-6	89	DRR 21-0966	7/7/21
B 3.5.5-1	21	DRR 05-0707	4/20/05
B 3.5.5-2	21	DRR 05-0707	4/20/05
B 3.5.5-3	89	DRR 21-0966	7/7/21
B 3.5.5-4	21	DRR 05-0707	4/20/05
TAB – B 3.6 CONTAINMENT SYSTEMS			
B 3.6.1-1	0	Amend. No. 123	12/18/99
B 3.6.1-2	81	DRR 19-1027	10/28/19
B 3.6.1-3	0	Amend. No. 123	12/18/99
B 3.6.1-4	96	N/A	12/11/2024
B 3.6.1-5	96	N/A	12/11/2024
B 3.6.2-1	81	DRR 19-1027	10/28/19
B 3.6.2-2	0	Amend. No. 123	12/18/99
B 3.6.2-3	0	Amend. No. 123	12/18/99
B 3.6.2-4	0	Amend. No. 123	12/18/99
B 3.6.2-5	0	Amend. No. 123	12/18/99
B 3.6.2-6	89	DRR 21-0966	7/7/21
B 3.6.2-7	89	DRR 21-0966	7/7/21
B 3.6.3-1	0	Amend. No. 123	12/18/99
B 3.6.3-2	84	DRR 20-0400	08/18/20
B 3.6.3-3	90	DRR 21-1229	11/3/21
B 3.6.3-4	49	DRR 11-0014	1/31/11
B 3.6.3-5	49	DRR 11-0014	1/31/11
B 3.6.3-6	49	DRR 11-0014	1/31/11
B 3.6.3-7	90	DRR 21-1229	11/3/21
B 3.6.3-8	36	DRR 08-0255	3/11/08
B 3.6.3-9	90	DRR 21-1299	11/3/21
B 3.6.3-10	89	DRR 21-0966	7/7/21
B 3.6.3-11	36	DRR 08-0255	3/11/08
B 3.6.3-12	89	DRR 21-0966	7/7/21

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TAB – B 3.6 CONTAINMENT SYSTEMS (continued)			
B 3.6.3-13	89	DRR 21-0966	7/7/21
B 3.6.3-14	36	DRR 08-0255	3/11/08
B 3.6.3-15	39	DRR 08-1096	8/28/08
B 3.6.3-16	39	DRR 08-1096	8/28/08
B 3.6.3-17	36	DRR 08-0255	3/11/08
B 3.6.3-18	36	DRR 08-0255	3/11/08
B 3.6.3-19	82	DRR 20-0077	1/29/20
B 3.6.4-1	39	DRR 08-1096	8/28/08
B 3.6.4-2	0	Amend. No. 123	12/18/99
B 3.6.4-3	89	DRR 21-0966	7/7/21
B 3.6.5-1	0	Amend. No. 123	12/18/99
B 3.6.5-2	37	DRR 08-0503	4/8/08
B 3.6.5-3	89	DRR 21-0966	7/7/21
B 3.6.5-4	0	Amend. No. 123	12/18/99
B 3.6.6-1	81	DRR 19-1027	10/28/19
B 3.6.6-2	63	DRR 14-1572	7/1/14
B 3.6.6-3	37	DRR 08-0503	4/8/08
B 3.6.6-4	81	DRR 19-1027	10/28/19
B 3.6.6-5	0	Amend. No. 123	12/18/99
B 3.6.6-6	18	DRR 04-1018	9/1/04
B 3.6.6-7	89	DRR 21-0966	7/7/21
B 3.6.6-8	89	DRR 21-0966	7/7/21
B 3.6.6-9	72	DRR 15-1918	10/26/15
B 3.6.6-10	89	DRR 21-0966	7/7/21
B 3.6.6.11	80	DRR 19-0524	5/30/19
B 3.6.7-1	0	Amend. No. 123	12/18/99
B 3.6.7-2	81	DRR 19-1027	10/28/19
B 3.6.7-3	89	DRR 21-0966	7/7/21
B 3.6.7-4	89	DRR 21-0966	7/7/21
B 3.6.7-5	89	DRR 21-0966	7/7/21
TAB – B 3.7 PLANT SYSTEMS			
B 3.7.1-1	0	Amend. No. 123	12/18/99
B 3.7.1-2	84	DRR 20-0400	08/18/20
B 3.7.1-3	0	Amend. No. 123	12/18/99
B 3.7.1-4	84	DRR 20-0400	08/18/20
B 3.7.1-5	84	DRR 20-0400	08/18/20
B 3.7.1-6	84	DRR 20-0400	08/18/20
B 3.7.2-1	44	DRR 09-1744	10/28/09
B 3.7.2-2	82	DRR 20-0077	1/29/20
B 3.7.2-3	82	DRR 20-0077	1/29/20
B 3.7.2-4	81	DRR 19-1027	10/28/19
B 3.7.2-5	82	DRR 20-0077	1/29/20
B 3.7.2-6	82	DRR 20-0077	1/29/20
B 3.7.2-7	82	DRR 20-0077	1/29/20
B 3.7.2-8	82	DRR 20-0077	1/29/20
B 3.7.2-9	89	DRR 21-0966	7/7/21
B 3.7.2-10	81	DRR 19-1027	10/28/19
B 3.7.2-11	44	DRR 09-1744	10/28/09
B 3.7.3-1	37	DRR 08-0503	4/8/08
B 3.7.3-2	50	DRR 11-0449	3/9/11

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TAB – B 3.7 PLANT SYSTEMS (continued)			
B 3.7.3-3	37	DRR 08-0503	4/8/08
B 3.7.3-4	37	DRR 08-0503	4/8/08
B 3.7.3-5	37	DRR 08-0503	4/8/08
B 3.7.3-6	37	DRR 08-0503	4/8/08
B 3.7.3-7	37	DRR 08-0503	4/8/08
B 3.7.3-8	37	DRR 08-0503	4/8/08
B 3.7.3-9	66	DRR 14-2329	11/6/14
B 3.7.3-10	89	DRR 21-0966	7/7/21
B 3.7.3-11	37	DRR 08-0503	4/8/08
B 3.7.4-1	1	DRR 99-1624	12/18/99
B 3.7.4-2	84	DRR 20-0400	08/18/20
B 3.7.4-3	19	DRR 04-1414	10/12/04
B 3.7.4-4	19	DRR 04-1414	10/12/04
B 3.7.4-5	89	DRR 21-0966	7/7/21
B 3.7.5-1	54	DRR 11-2394	11/16/11
B 3.7.5-2	54	DRR 11-2394	11/16/11
B 3.7.5-3	0	Amend. No. 123	12/18/99
B 3.7.5-4	85	DRR 20-0988	10/24/20
B 3.7.5-5	76	DRR 17-0343	2/21/17
B 3.7.5-6	85	DRR 20-0988	10/24/20
B 3.7.5-7	90	DRR 21-1229	11/3/21
B 3.7.5-8	90	DRR 21-1229	11/3/21
B 3.7.5-9	90	DRR 21-1229	11/3/21
B 3.7.5-10	90	DRR 21-1229	11/3/21
B 3.7.6-1	0	Amend. No. 123	12/18/99
B 3.7.6-2	0	Amend. No. 123	12/18/99
B 3.7.6-3	89	DRR 21-0966	7/7/21
B 3.7.7-1	0	Amend. No. 123	12/18/99
B 3.7.7-2	77	DRR 17-1001	6/22/17
B 3.7.7-3	92	DRR 22-0767	11/3/22
B 3.7.7-4	89	DRR 21-0966	7/7/21
B 3.7.8-1	0	Amend. No. 123	12/18/99
B 3.7.8-2	0	Amend. No. 123	12/18/99
B 3.7.8-3	0	Amend. No. 123	12/18/99
B 3.7.8-4	89	DRR 21-0966	7/7/21
B 3.7.8-5	89	DRR 21-0966	7/7/21
B 3.7.9-1	3	Amend. No. 134	7/14/00
B 3.7.9-2	3	Amend. No. 134	7/14/00
B 3.7.9-3	89	DRR 21-0966	7/7/21
B 3.7.9-4	3	Amend. No. 134	7/14/00
B 3.7.10-1	64	DRR 14-1822	8/28/14
B 3.7.10-2	81	DRR 19-1027	10/28/19
B 3.7.10-3	81	DRR 19-1027	10/28/19
B 3.7.10-4	81	DRR 19-1027	10/28/19
B 3.7.10-5	81	DRR 19-1027	10/28/19
B 3.7.10-6	57	DRR 13-0006	1/16/13
B 3.7.10-7	89	DRR 21-0966	7/7/21
B 3.7.10-8	89	DRR 21-0966	7/7/21
B 3.7.10-9	81	DRR 19-1027	10/28/19
B 3.7.11-1	0	Amend. No. 123	12/18/99
B 3.7.11-2	57	DRR 13-0006	1/16/13

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TAB – B 3.7 PLANT SYSTEMS (continued)			
B 3.7.11-3	89	DRR 21-0966	7/7/21
B 3.7.11-4	63	DRR 14-1572	7/1/14
B 3.7.12-1	0	Amend. No. 123	12/18/99
B 3.7.13-1	24	DRR 06-0051	2/28/06
B 3.7.13-2	81	DRR 19-1027	10/28/19
B 3.7.13-3	81	DRR 19-1027	10/28/19
B 3.7.13-4	81	DRR 19-1027	10/28/19
B 3.7.13-5	96	N/A	12/11/2024
B 3.7.13-6	89	DRR 21-0966	7/7/21
B 3.7.13-7	89	DRR 21-0966	7/7/21
B 3.7.13-8	89	DRR 21-0966	7/7/21
B 3.7.14-1	0	Amend. No. 123	12/18/99
B 3.7.15-1	81	DRR 19-1027	10/28/19
B 3.7.15-2	89	DRR 21-0966	7/7/21
B 3.7.15-3	81	DRR 19-1027	10/28/19
B 3.7.16-1	5	DRR 00-1427	10/12/00
B 3.7.16-2	95	N/A	4/25/24
B 3.7.16-3	89	DRR 21-0966	7/7/21
B 3.7.17-1	7	DRR 01-0474	5/1/01
B 3.7.17-2	7	DRR 01-0474	5/1/01
B 3.7.17-3	5	DRR 00-1427	10/12/00
B 3.7.18-1	81	DRR 19-1027	10/28/19
B 3.7.18-2	81	DRR 19-1027	10/28/19
B 3.7.18-3	89	DRR 21-0966	7/7/21
B 3.7.19-1	44	DRR 09-1744	10/28/09
B 3.7.19-2	54	DRR 11-2394	11/16/11
B 3.7.19-3	54	DRR 11-2394	11/16/11
B 3.7.19-4	61	DRR 14-0346	2/27/14
B 3.7.19-5	61	DRR 14-0346	2/27/14
B 3.7.19-6	89	DRR 21-0966	7/7/21
B 3.7.19-7	89	DRR 21-0966	7/7/21
B 3.7.20-1	79	DRR 18-1579	10/22/18
B 3.7.20-2	90	DRR 21-1229	11/3/21
B 3.7.20-3	85	DRR 20-0988	10/24/20
B 3.7.20-4	89	DRR 21-0966	7/7/21
B 3.7.20-5	89	DRR 21-0966	7/7/21
TAB – B 3.8 ELECTRICAL POWER SYSTEMS			
B 3.8.1-1	98	N/A	10/15/25
B 3.8.1-2	93	DRR 22-1363	2/2/23
B 3.8.1-3	95	N/A	4/25/24
B 3.8.1-4	96	N/A	12/11/2024
B 3.8.1-5	93	DRR 22-1363	2/2/23
B 3.8.1-6	93	DRR 22-1363	2/2/23
B 3.8.1-7	93	DRR 22-1363	2/2/23
B 3.8.1-8	93	DRR 22-1363	2/2/23
B 3.8.1-9	93	DRR 22-1363	2/2/23
B 3.8.1-10	93	DRR 22-1363	2/2/23
B 3.8.1-11	93	DRR 22-1363	2/2/23
B 3.8.1-12	93	DRR 22-1363	2/2/23
B 3.8.1-13	93	DRR 22-1363	2/2/23

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TAB – B 3.8 ELECTRICAL POWER SYSTEMS (continued)			
B 3.8.1-14	93	DRR 22-1363	2/2/23
B 3.8.1-15	47	DRR 10-1089	6/16/10
B 3.8.1-16	26	DRR 06-1350	7/24/06
B 3.8.1-17	93	DRR 22-1363	2/2/23
B 3.8.1-18	95	N/A	4/25/24
B 3.8.1-19	89	DRR 21-0966	7/7/21
B 3.8.1-20	89	DRR 21-0966	7/7/21
B 3.8.1-21	93	DRR 22-1363	2/2/23
B 3.8.1-22	89	DRR 21-0966	7/7/21
B 3.8.1-23	93	DRR 22-1363	2/2/23
B 3.8.1-24	74	DRR 16-1182	7/7/16
B 3.8.1-25	89	DRR 21-0966	7/7/21
B 3.8.1-26	89	DRR 21-0966	7/7/21
B 3.8.1-27	89	DRR 21-0966	7/7/21
B 3.8.1-28	93	DRR 22-1363	2/2/23
B 3.8.1-29	96	N/A	12/11/2024
B 3.8.1-30	96	N/A	12/11/2024
B 3.8.1-31	89	DRR 21-0966	7/7/21
B 3.8.1-32	89	DRR 21-0966	7/7/21
B 3.8.1-33	89	DRR 21-0966	7/7/21
B 3.8.1-34	93	DRR 22-1363	2/2/23
B 3.8.2-1	57	DRR 13-0006	1/16/13
B 3.8.2-2	0	Amend. No. 123	12/18/99
B 3.8.2-3	92	DRR 22-0767	11/3/22
B 3.8.2-4	92	DRR 22-0767	11/3/22
B 3.8.2-5	57	DRR 13-0006	1/16/13
B 3.8.2-6	57	DRR 13-0006	1/16/13
B 3.8.2-7	57	DRR 13-0006	1/16/13
B 3.8.3-1	1	DRR 99-1624	12/18/99
B 3.8.3-2	90	DRR 21-1229	11/3/21
B 3.8.3-3	0	Amend. No. 123	12/18/99
B 3.8.3-4	1	DRR 99-1624	12/18/99
B 3.8.3-5	0	Amend. No. 123	12/18/99
B 3.8.3-6	89	DRR 21-0966	7/7/21
B 3.8.3-7	12	DRR 02-1062	9/26/02
B 3.8.3-8	89	DRR 21-0966	7/7/21
B 3.8.3-9	0	Amend. No. 123	12/18/99
B 3.8.4-1	0	Amend. No. 123	12/18/99
B 3.8.4-2	0	Amend. No. 123	12/18/99
B 3.8.4-3	93	DRR 22-1363	2/2/23
B 3.8.4-4	0	Amend. No. 123	12/18/99
B 3.8.4-5	93	DRR 22-1363	2/2/23
B 3.8.4-6	89	DRR 21-0966	7/7/21
B 3.8.4-7	6	DRR 00-1541	3/13/01
B 3.8.4-8	93	DRR 22-1363	2/2/23
B 3.8.4-9	89	DRR 21-0966	7/7/21
B 3.8.5-1	57	DRR 13-0006	1/16/13
B 3.8.5-2	95	N/A	4/25/24
B 3.8.5-3	57	DRR 13-0006	1/16/13
B 3.8.5-4	57	DRR 13-0006	1/16/13
B 3.8.5-5	57	DRR 13-0006	1/16/13

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TAB – B 3.8 ELECTRICAL POWER SYSTEMS (continued)			
B 3.8.6-1	0	Amend. No. 123	12/18/99
B 3.8.6-2	0	Amend. No. 123	12/18/99
B 3.8.6-3	89	DRR 21-0966	7/7/21
B 3.8.6-4	89	DRR 21-0966	7/7/21
B 3.8.6-5	0	Amend. No. 123	12/18/99
B 3.8.6-6	0	Amend. No. 123	12/18/99
B 3.8.7-1	69	DRR 15-0493	3/26/15
B 3.8.7-2	69	DRR 15-0493	3/26/15
B 3.8.7-3	89	DRR 21-0966	7/7/21
B 3.8.7-4	0	Amend. No. 123	12/18/99
B 3.8.8-1	57	DRR 13-0006	1/16/13
B 3.8.8-2	0	Amend. No. 123	12/18/99
B 3.8.8-3	69	DRR 15-0493	3/26/15
B 3.8.8-4	57	DRR 13-0006	1/16/13
B 3.8.8-5	93	DRR 22-1363	2/2/23
B 3.8.9-1	54	DRR 11-2394	11/16/11
B 3.8.9-2	69	DRR 15-0493	3/26/15
B 3.8.9-3	54	DRR 11-2394	11/16/11
B 3.8.9-4	0	Amend. No. 123	12/18/99
B 3.8.9-5	69	DRR 15-0493	3/26/15
B 3.8.9-6	0	Amend. No. 123	12/18/99
B 3.8.9-7	0	Amend. No. 123	12/18/99
B 3.8.9-8	89	DRR 21-0966	7/7/21
B 3.8.9-9	0	Amend. No. 123	12/18/99
B 3.8.10-1	57	DRR 13-0006	1/16/13
B 3.8.10-2	0	Amend. No. 123	12/18/99
B 3.8.10-3	0	Amend. No. 123	12/18/99
B 3.8.10-4	57	DRR 13-0006	1/16/13
B 3.8.10-5	57	DRR 13-0006	1/16/13
B 3.8.10-6	89	DRR 21-0966	7/7/21
TAB – B 3.9 REFUELING OPERATIONS			
B 3.9.1-1	0	Amend. No. 123	12/18/99
B 3.9.1-2	19	DRR 04-1414	10/12/04
B 3.9.1-3	19	DRR 04-1414	10/12/04
B 3.9.1-4	89	DRR 21-0966	7/7/21
B 3.9.2-1	0	Amend. No. 123	12/18/99
B 3.9.2-2	0	Amend. No. 123	12/18/99
B 3.9.2-3	89	DRR 21-0966	7/7/21
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Note 1 The page number is listed on the center of the bottom of each page.

Note 2 The revision number is listed in the lower right hand corner of each page. The Revision number will be page specific.

Note 3 The change document will be the document requesting the change. Amendment No. 123 issued the improved Technical Specifications and associated Bases which affected each page. The NRC has indicated that Bases changes will not be issued with License Amendments. As of Revision 94, the TS Bases is no longer controlled in PMAC. Therefore, starting with Revision 94, the DRR number will be N/A.

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B 3.2 POWER DISTRIBUTION LIMITS

B 3.2.1 Heat Flux Hot Channel Factor ($F_Q(Z)$) (RAOC - T(Z) Methodology)

BASES

BACKGROUND

The purpose of the limits on the values of $F_Q(Z)$ is to limit the local (i.e., pellet) peak power density. The value of $F_Q(Z)$ varies along the axial height (Z) of the core.

$F_Q(Z)$ is defined as the maximum local fuel rod linear power density divided by the average fuel rod linear power density, assuming nominal fuel pellet and fuel rod dimensions. Therefore, $F_Q(Z)$ is a measure of the peak fuel pellet power within the reactor core.

During power operation, the global power distribution is limited by LCO 3.2.3, "AXIAL FLUX DIFFERENCE (AFD)," and LCO 3.2.4, "QUADRANT POWER TILT RATIO (QPTR)," which are directly and continuously measured process variables. These LCOs, along with LCO 3.1.4, “Rod Group Alignment Limits,” LCO 3.1.5, “Shutdown Bank Insertion Limits,” and LCO 3.1.6, "Control Bank Insertion Limits," maintain the core limits on power distributions on a continuous basis.

$F_Q(Z)$ varies with fuel loading patterns, control bank insertion, fuel burnup, and changes in axial power distribution.

$F_Q(Z)$ is not directly measurable but is inferred from a power distribution measurement obtained with either the movable incore detector system or the Power Distribution Monitoring System (PDMS). The results of the three-dimensional power distribution measurement are analyzed to derive a measured value for $F_Q(Z)$. These measurements are generally taken with the core at or near equilibrium conditions. However, because this value represents an equilibrium condition, it does not include the variations in the value of $F_Q(Z)$ that are present during nonequilibrium situations, such as load following.

To account for these possible variations, the elevation dependent measured planar radial peaking factors, $F_{XY}(Z)$, are increased by an elevation dependent factor, $[T(Z)]^{COLR}$, that accounts for the expected maximum values of the transient axial power shapes postulated to occur during RAOC operation. Thus, $[T(Z)]^{COLR}$ accounts for the worst case non-equilibrium power shapes that are expected for the assumed RAOC operating space.

The RAOC operating space is defined as the combination of AFD and Control Bank Insertion Limits assumed in the calculation of a particular $[T(Z)]^{COLR}$ function. The $[T(Z)]^{COLR}$ factors are directly dependent on the

BASES

BACKGROUND (continued)

AFD and Control Bank Insertion Limit assumptions. The COLR may contain different $[T(Z)]^{COLR}$ functions that reflect different operating space assumptions. If the limit on $F_Q(Z)$ is exceeded, a more restrictive operating space may be implemented to gain margin for future non-equilibrium operation.

Core monitoring and control under nonsteady state conditions are accomplished by operating the core within the limits of the appropriate LCOs, including the limits on AFD, QPTR, and control rod insertion.

APPLICABLE SAFETY ANALYSES

This LCO precludes core power distributions that violate the following fuel design criteria:

a. During a large break loss of coolant accident (LOCA), the peak cladding temperature must not exceed 2200°F (Ref. 1);

b. During a loss of forced reactor coolant flow accident, there must be at least 95% probability at the 95% confidence level (the 95/95 DNB criterion) that the hot fuel rod in the core does not experience a departure from nucleate boiling (DNB) condition;

c. During an ejected rod accident, the average fuel pellet enthalpy at the hot spot in irradiated fuel must not exceed 200 cal/gm (Ref. 2); and

d. The control rods must be capable of shutting down the reactor with a minimum required SDM with the highest worth control rod stuck fully withdrawn (Ref. 3).

Limits on $F_Q(Z)$ ensure that the value of the initial total peaking factor assumed in the accident analyses remains valid. Other criteria must also be met (e.g., maximum cladding oxidation, maximum hydrogen generation, coolable geometry, and long term cooling). However, the LOCA peak cladding temperature is typically most limiting.

$F_Q(Z)$ limits assumed in the LOCA analysis are typically limiting relative to (i.e., lower than) the $F_Q(Z)$ limit assumed in safety analyses for other postulated accidents. Therefore, this LCO provides conservative limits for other postulated accidents.

$F_Q(Z)$ satisfies Criterion 2 of 10 CFR 50.36(c)(2)(ii).

BASES

LCO

The Heat Flux Hot Channel Factor, $F_Q(Z)$, shall be limited by the following relationships:

$$F_Q(Z) \leq \frac{CFQ}{P} K(Z) \qquad \text{for } P > 0.5$$

$$F_Q(Z) \leq \frac{CFQ}{0.5} K(Z) \qquad \text{for } P \leq 0.5$$

where: CFQ = F_Q^{RTP} is the $F_Q(Z)$ limit at RTP provided in the COLR, K(Z) is the normalized $F_Q(Z)$ limit as a function of core height provided in the COLR, and

$$P = \frac{\text{THERMAL POWER}}{\text{RTP}}$$

The actual values of CFQ and K(Z) are given in the COLR.

For Relaxed Axial Offset Control operation, $F_Q(Z)$ is approximated by $F_Q^C(Z)$ and $F_Q^W(Z)$. Thus, both $F_Q^C(Z)$ and $F_Q^W(Z)$ must meet the preceding limits on $F_Q(Z)$.

An $F_Q^C(Z)$ evaluation requires obtaining a power distribution measurement in MODE 1, from which we obtain the measured value ($F_Q^M(Z)$) of $F_Q(Z)$.

If the power distribution measurement is obtained with the movable incore detector system,

$$F_Q^C(Z) = F_Q^M(Z)\ (1.03)\ (1.05) = F_Q^M(Z)\ (1.0815)$$

where 1.03 is a factor that accounts for fuel manufacturing tolerances and 1.05 is a factor that accounts for flux map measurement uncertainty. (Ref. 4)

If the power distribution measurement is obtained with the Power Distribution Monitoring System,

$$F_Q^C(Z) = F_Q^M(Z)\ (1.03)\ (1.00 + U_Q/100)$$

where 1.03 is a factor that accounts for fuel manufacturing tolerances and U_Q is a factor that accounts for Power Distribution Monitoring System measurement uncertainty (%), determined as described in Reference 6.

$F_Q^C(Z)$ is an excellent approximation for $F_Q(Z)$ when the reactor is at the steady state power at which the power distribution measurement was taken.

BASES

LCO (continued)

The expression for $F_Q^W(Z)$ is:

$$F_Q^W(Z)=F_{XY}^M(Z)\frac{[T(Z)]^{COLR}}{P}A_{XY}(Z)Rj[1.0815] \quad \text{for } P>0.5$$

$$F_Q^W(Z)=F_{XY}^M(Z)\frac{[T(Z)]^{COLR}}{0.5}A_{XY}(Z)Rj[1.0815] \quad \text{for } P\leq 0.5$$

The various factors in this expression are defined below:

$F_{XY}^M(Z)$ is the measured radial peaking factor at axial location Z and is equal to the value of $F_Q^M(Z)/P^M(Z)$, where $P^M(Z)$ is the measured core average axial power shape.

$[T(Z)]^{COLR}$ is the cycle and burnup dependent function, specified in the COLR, which accounts for power distribution transients encountered during non-equilibrium normal operation. $[T(Z)]^{COLR}$ functions are specified for each analyzed RAOC operating space (i.e., each unique combination of AFD limits and Control Bank Insertion Limits). The $[T(Z)]^{COLR}$ functions account for the limiting non-equilibrium axial power shapes postulated to occur during normal operation for each RAOC operating space. Limiting power shapes at both full and reduced power operation are considered in determining the maximum values of $[T(Z)]^{COLR}$. The $[T(Z)]^{COLR}$ functions also account for the following effects: (1) the presence of spacer grids in the fuel assembly, (2) the increase in radial peaking in rodded core planes due to the presence of control rods during non-equilibrium normal operation, (3) the increase in radial peaking that occurs during part-power operation due to reduced fuel and moderator temperatures, and (4) the increase in radial peaking due to non-equilibrium xenon effects. The $[T(Z)]^{COLR}$ functions are normally calculated assuming that the Surveillance is performed at nominal RTP conditions with all shutdown and control rods full withdrawn, i.e., all rods out (ARO) Surveillance specific $[T(Z)]^{COLR}$ values may be generated for a given surveillance core condition.

P is the THERMAL POWER / RTP.

$A_{XY}(Z)$ is a function that adjusts the $F_Q^W(Z)$ Surveillance for differences between the reference core condition assumed in generating the $[T(Z)]^{COLR}$ function and the actual core condition that exists when the Surveillance is performed. Normally, this reference core condition is 100% RTP, all rods out, and equilibrium xenon. For simplicity, $A_{XY}(Z)$ may be assumed to be 1.0, as this will typically result in an accurate $F_Q^W(Z)$ Surveillance result for a Surveillance that is performed at or near the reference core condition, and an underestimation of the available margin to the F_Q limit for Surveillances that are performed at core conditions different from the reference condition. Alternately, the $A_{XY}(Z)$

BASES

LCO (continued)

function may be calculated using the NRC approved methodology in Reference 7.

1.0815 is a factor that accounts for fuel manufacturing tolerances and measurement uncertainty.

Rj is a cycle and burnup dependent analytical factor specified in the COLR that accounts for potential increases in $F_Q^W(Z)$ between Surveillances. Rj values are provided for each RAOC operating space.

The $F_Q(Z)$ limits define limiting values for core power peaking that precludes peak cladding temperatures above 2200°F during either a large or small break LOCA.

This LCO requires operation within the bounds assumed in the safety analyses. Violating the LCO limits for $F_Q(Z)$ could result in unacceptable consequences if a design basis event were to occur while $F_Q(Z)$ exceeds its specified limits. Calculations are performed in the core design process to confirm that the core can be controlled in such a manner during operation that it can stay within the LOCA $F_Q(Z)$ limits. If $F_Q(Z)$ cannot be maintained within the LCO limits, a more restrictive RAOC operating space must be implemented or core power limits and AFD limits must be reduced.

APPLICABILITY

The $F_Q(Z)$ limits must be maintained in MODE 1 to prevent core power distributions from exceeding the limits assumed in the safety analyses. Applicability in other MODES is not required because there is either insufficient stored energy in the fuel or insufficient energy being transferred to the reactor coolant to require a limit on the distribution of core power.

ACTIONS

A.1

Reducing THERMAL POWER by ≥ 1% RTP for each 1% by which $F_Q^C(Z)$ exceeds its limit, maintains an acceptable absolute power density. $F_Q^C(Z)$ is $F_Q^M(Z)$ multiplied by factors which account for manufacturing tolerances and measurement uncertainties. $F_Q^M(Z)$ is the measured value of $F_Q(Z)$. The Completion Time of 15 minutes provides an acceptable time to reduce power in an orderly manner and without allowing the plant to remain in an unacceptable condition for an extended period of time. The maximum allowable power level initially determined by Required Action A.1 may be affected by subsequent determinations of $F_Q^C(Z)$ and would require power reductions within 15 minutes of the $F_Q^C(Z)$ determination, if necessary to comply with the decreased maximum allowable power level.

BASES

ACTIONS

A.1 (continued)

Decreases in $F_Q^C(Z)$ would allow increasing the maximum allowable power level and increasing power up to this revised limit.

If an F_Q surveillance is performed at 100% RTP conditions, and both $F_Q^C(Z)$ and $F_Q^W(Z)$ exceed their limits, the option to reduce the THERMAL POWER limit in accordance with Required Action B.2.1 instead of implementing a new operating space in accordance with Required Action B.1.1, will result in a further power reduction after Required Action A.1 has been completed. However, this further power reduction would be permitted to occur over the next 4 hours. In the event the evaluated THERMAL POWER reduction in the COLR for Required Action B.2.1 did not result in a further power reduction (for example, if both Condition A and Condition B were entered at less than 100% RTP conditions), then the THERMAL POWER level established as a result of completing Required Action A.1 will take precedence, and will establish the effective operating power level limit for the unit until both Conditions A and B are exited.

A.2

A reduction of the Power Range Neutron Flux - High trip setpoints by ≥ 1% for each 1% that THERMAL POWER is limited below RTP by Required Action A.1, is a conservative action for protection against the consequences of severe transients with unanalyzed power distributions. The Completion Time of 72 hours is sufficient considering the small likelihood of a severe transient in this time period and the preceding prompt reduction in THERMAL POWER in accordance with Required Action A.1. The maximum allowable Power Range Neutron Flux - High trip setpoints initially determined by Required Action A.2 may be affected by subsequent determinations of $F_Q^C(Z)$ and would require Power Range Neutron Flux - High trip setpoint reductions within 72 hours of $F_Q^C(Z)$ determination, if necessary to comply with the decreased maximum allowable Power Range Neutron Flux - High trip setpoints.

A.3

Reduction in the Overpower ΔT trip setpoints by ≥ 1% for each 1% that THERMAL POWER is limited below RTP by Required Action A.1, is a conservative action for protection against the consequences of severe transients with unanalyzed power distributions. The Completion Time of 72 hours is sufficient considering the small likelihood of a severe transient in this time period, and the preceding prompt reduction in THERMAL POWER in accordance with Required Action A.1. The maximum allowable Overpower ΔT trip setpoints initially determined by Required Action A.3 may be affected by subsequent determinations of $F_Q^C(Z)$ and

BASES

ACTIONS

A. 3 (continued)

would require Overpower ΔT trip setpoint reductions within 72 hours of the $F_Q^C(Z)$ determination, if necessary to comply with the decreased maximum allowable Overpower ΔT trip setpoints. Decreases in $F_Q^C(Z)$ would allow increasing the maximum Overpower ΔT trip setpoints.

A.4

Verification that $F_Q^C(Z)$ has been restored to within its limit, by performing SR 3.2.1.1 and SR 3.2.1.2 prior to increasing THERMAL POWER above the limit imposed by Required Action A.1, ensures that core conditions during operation at higher power levels and future operation are consistent with safety analyses assumptions. Inherent in this action is identification of the cause of the out of limit condition and the correction of the cause to the extent necessary to allow safe operation at the higher power level.

Condition A is modified by a Note that requires Required Action A.4 to be performed whenever the Condition is entered prior to increasing THERMAL POWER above the limit of Required Action A.1. The Note also states that SR 3.2.1.2 is not required to be performed if this Condition is entered prior to THERMAL POWER exceeding 75% RTP after a refueling. This ensures that SR 3.2.1.1 and SR 3.2.1.2 (if required) will be performed prior to increasing THERMAL POWER above the limit of Required Action A.1, even when Condition A is exited prior to performing Required Action A.4. Performance of SR 3.2.1.1 and SR 3.2.1.2 are necessary to assure $F_Q(Z)$ is properly evaluated prior to increasing THERMAL POWER.

B.1.1

If it is found that the maximum calculated value of $F_Q(Z)$ that can occur during normal maneuvers, $F_Q^W(Z)$, exceeds its specified limits, there exists a potential for $F_Q^C(Z)$ to become excessively high if a normal operational transient occurs. Implementing a more restrictive RAOC operating space, as specified in the COLR, within the allowed Completion Time of 4 hours will restrict the AFD such that peaking factor limits will not be exceeded during non-equilibrium normal operation. Several RAOC operating spaces, representing successively smaller AFD envelopes and, optionally, shallower Control Bank Insertion Limits, may be specified in the COLR. The corresponding T(Z) functions for these operating spaces can be used to determine which RAOC operating space will result in acceptable non-equilibrium operation within the $F_Q^W(Z)$ limit.

BASES

ACTIONS (continued)

B.1.2

If it is found that the maximum calculated value of $F_Q(Z)$ that can occur during normal maneuvers, $F_Q^W(Z)$, exceeds its specified limits, there exists a potential for $F_Q^C(Z)$ to become excessively high if a normal operational transient occurs. As discussed above, Required Action B.1.1 requires that a new RAOC operating space be implemented to restore $F_Q^W(Z)$ to within its limits. Required Action B.1.2 requires that SR 3.2.1.1 and SR 3.2.1.2 be performed if control rod motion occurs as a result of implementing the new RAOC operating space in accordance with Required Action B.1.1. The performance of SR 3.2.1.1 and SR 3.2.1.2 is necessary to assure $F_Q(Z)$ is properly evaluated after any rod motion resulting from the implementation of a new RAOC operating space in accordance with Required Action B.1.1.

B.2.1

When $F_Q^W(Z)$ exceeds it limit, Required Action B.2.1 may be implemented instead of Required Action B.1. Required Action B.2.1 limits THERMAL POWER to less than RTP by the amount specified in the COLR. It also requires reductions in the AFD limits by the amount specified in the COLR. This maintains an acceptable absolute power density relative to the maximum power density value assumed in the safety analyses.

If the required $F_Q^W(Z)$ margin improvement exceeds the margin improvement available from the pre-analyzed THERMAL POWER and AFD reductions provided in the COLR, then THERMAL POWER must be further reduced to less than or equal to 50% RTP. In this case, reducing THERMAL POWER to less than or equal to 50% RTP will provide additional margin in the transient F_Q by the required change in THERMAL POWER and the increase in the F_Q limit. This will ensure that the F_Q limit is met during transient operation that may occur at or below 50% RTP.

The Completion Time of 4 hours provides an acceptable time to reduce the THERMAL POWER and AFD limits in an orderly manner to preclude entering an unacceptable condition during future non-equilibrium operation. The limit on THERMAL POWER initially determined by Required Action B.2.1 may be affected by subsequent determinations of $F_Q^W(Z)$ that are not within limit and could require power reductions within 4 hours of the of the subsequent $F_Q^W(Z)$ determination, if necessary, to comply with the decreased THERMAL POWER limit. In short, the 4 hour Completion Time for Required Action B.2.1 applies after each $F_Q^W(Z)$ determination. Decreases in subsequent $F_Q^W(Z)$ measurements while in Condition B would allow increasing the THERMAL POWER limit and increasing THERMAL POWER up to this revised limit.

BASES

ACTIONS

B.2.1 (continued)

Required Action B.2.1 is modified by a Note that states Required Action B.2.4 shall be completed whenever Required Action B.2.1 is performed prior to increasing THERMAL POWER above the limit of Required Action B.2.1. Required Action B.2.4 requires the performance of SR 3.2.1.1 and SR 3.2.1.2 prior to increasing THERMAL POWER above the limit established by Required Action B.2.1. The Note ensures that the SRs will be performed even if Condition B may be exited prior to performing Required Action B.2.4. The performance of SR 3.2.1.1 and SR 3.2.1.2 is necessary to ensure $F_Q(Z)$ is properly evaluated prior to increasing THERMAL POWER.

If an F_Q surveillance is performed at 100% RTP conditions, and both $F_Q^C(Z)$ and $F_Q^W(Z)$ exceed their limits, the option to reduce the THERMAL POWER limit in accordance with Required Action B.2.1 instead of implementing a new operating space in accordance with Required Action B.1, will result in a further power reduction after Required Action A.1 has been completed. However, this further power reduction would be permitted to occur over the next 4 hours. In the event the evaluated THERMAL POWER reduction in the COLR for Required Action B.2.1 did not result in a further power reduction (for example, if both Condition A and Condition B were entered at less than 100% RTP conditions), then the THERMAL POWER level established as a result of completing Required Action A.1 will take precedence, and will establish the effective operating power level limit for the unit until both Conditions A and B are exited.

B.2.2

A reduction of the Power Range Neutron Flux - High trip setpoints by ≥ 1% for each 1% by which the maximum allowable power is reduced is a conservative action for protection against the consequences of severe transients with unanalyzed power distributions. The Completion Time of 72 hours is sufficient considering the small likelihood of a severe transient in this time period and the preceding prompt reduction in the THERMAL POWER limit and AFD limits in accordance with Required Action B.2.1. The maximum allowable Power Range Neutron Flux - High trip setpoints initially determined by Required Action B.2.2 may be affected by subsequent determinations of $F_Q^W(Z)$ that are not within limit and could require Power Range Neutron Flux - High trip setpoint reductions within 72 hours of the subsequent $F_Q^W(Z)$ determination, if necessary, to comply with the decreased maximum allowable Power Range Neutron Flux - High trip setpoints. In short, the 72-hour Completion Time for Required Action B.2.2 applies after each $F_Q^W(Z)$ determination.

BASES

ACTIONS (continued)

B.2.3

Reduction in the Overpower ΔT trip setpoints value of K_4 by ≥ 1% for each 1% by which the maximum allowable power is reduced is a conservative action for protection against the consequences of severe transients with unanalyzed power distributions. The Completion Time of 72 hours is sufficient considering the small likelihood of a severe transient in this time period, and the preceding prompt reduction in the THERMAL POWER limit and AFD limits in accordance with the Required Action B.2.1. The maximum allowable Overpower ΔT trip setpoints initially determined by Required Action B.2.3 may be affected by subsequent determinations of $F_Q^W(Z)$ that are not within limit and could require Overpower ΔT trip setpoint reductions within 72 hours of the subsequent $F_Q^W(Z)$ determination, if necessary, to comply with the decreased maximum allowable Overpower ΔT trip setpoints. In short, the 72 hour Completion Time for Required Action B.2.3 applies after each $F_Q^W(Z)$ determination. Decreases in subsequent $F_Q^W(Z)$ measurements while in Condition B would allow increasing the maximum allowable Overpower ΔT trip setpoints.

B.2.4

Verification that $F_Q^W(Z)$ has been restored to within its limit, by performing SR 3.2.1.1 and SR 3.2.1.2 prior to increasing THERMAL POWER above the maximum allowable power limit imposed by Required Action B.2.1, ensures that core conditions during operation at higher power levels and future operation are consistent with safety analyses assumptions.

C.1

If Required Actions A.1 through A.4 or B.1.1 through B.2.4 are not met within their associated Completion Times, the plant must be placed in a mode or condition in which the LCO requirements are not applicable. This is done by placing the plant in at least MODE 2 within 6 hours.

This allowed Completion Time is reasonable based on operating experience regarding the amount of time it takes to reach MODE 2 from full power operation in an orderly manner and without challenging plant systems.

BASES

SURVEILLANCE REQUIREMENTS

SR 3.2.1.1

Verification that $F_Q^C(Z)$ is within its specified limits involves increasing $F_Q^M(Z)$ to allow for manufacturing tolerance and measurement uncertainties in order to obtain $F_Q^C(Z)$, as described in the preceeding LCO section.

The limit with which $F_Q^C(Z)$ is compared varies inversely with power above 50% RTP and directly with a function called K(Z) provided in the COLR.

Performing this Surveillance in MODE 1 prior to exceeding 75% RTP following a refueling ensures that some determination of $F_Q^C(Z)$ is made prior to achieving a significant power level where the peak linear heat rate could approach the limits assumed in the safety analyses.

If THERMAL POWER has been increased by ≥ 10% RTP since the initial or most recent determination of $F_Q^C(Z)$, another evaluation of this factor is required within 24 hours after achieving equilibrium conditions at this higher power level (to ensure that $F_Q^C(Z)$ values are being reduced sufficiently with power increase to stay within the LCO limits).

Equilibrium conditions are achieved when the core is sufficiently stable at the intended operating conditions required to perform the Surveillance.

The allowance of up to 24 hours after achieving equilibrium conditions at the increased THERMAL POWER level to complete the next $F_Q^C(Z)$ surveillance applies to situations where the $F_Q^C(Z)$ has already been measured at least once at a reduced THERMAL POWER level. The observed margin in the previous Surveillance will provide assurance that increasing power up to the next plateau will not exceed the F_Q limit, and that the core is behaving as designed.

This Frequency condition is not intended to require verification of these parameters after every 10% increase in RTP above the THERMAL POWER at which the last verification was performed. It only requires verification after a THERMAL POWER is achieved for extended operation that is 10% higher than the THERMAL POWER at which $F_Q^C(Z)$ was last measured.

The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

BASES

SURVEILLANCE REQUIREMENTS (continued)

SR 3.2.1.2

The nuclear design process includes calculations performed to determine that the core can be operated within the $F_Q(Z)$ limits. Because power distribution measurements are taken at or near equilibrium conditions, the variations in power distribution resulting from normal operational maneuvers are not present in the measurements. These variations are, however, conservatively calculated by considering a wide range of unit maneuvers in normal operation.

The measured $F_Q(Z)$ can be determined through a synthesis of the measured planar radial peaking factors, $F_{XY}^M(Z)$, and the measured core average axial power shape, $P^M(Z)$. Thus, $F_Q^C(Z)$ is given by the following expression:

$$F_Q^C(Z) = F_{XY}^M(Z)\ P^M(Z)[1.0815] = F_Q^M\ (Z)[1.0815]$$

For RAOC operation, the analytical $[T(Z)]^{COLR}$ functions, specified in the COLR for each RAOC operating space, are used together with the measured $F_{XY}(Z)$ values to estimate $F_Q(Z)$ for non-equilibrium operation within the RAOC operating space. When the $F_{XY}(Z)$ values are measured at HFP ARO conditions ($A_{XY}(Z)$ equals 1.0), $F_Q^W(Z)$ is given by the following expression:

$$F_Q^W(Z) = F_{XY}^M(Z)\ [T(Z)]^{COLR}\ Rj\ [1.0815]$$

Non-equilibrium operation can result in significant changes to the axial power shape. To a lesser extent, non-equilibrium operation can increase the radial peaking factors, $F_{XY}(Z)$, through control rod insertion and through reduced Doppler and moderator feedback at part-power conditions.

The $[T(Z)]^{COLR}$ functions quantify these effects for the range of power shapes, control rod insertion, and power levels characteristic of the operating space. Multiplying $[T(Z)]^{COLR}$ by the measured full power, unrodded $F_{XY}^M(Z)$ value, and the factor that accounts for manufacturing and measurement uncertainties gives $F_Q^W(Z)$, the maximum total peaking factor postulated for non-equilibrium RAOC operation.

The limit with which $F_Q^W(Z)$ is compared varies inversely with power above 50% RTP and directly with the function K(Z) provided in the COLR.

BASES

SURVEILLANCE REQUIREMENTS

SR 3.2.1.2 (continued)

The $[T(Z)]^{COLR}$ functions are specified in the COLR for discrete core elevations. Flux map data are typically taken for 30 to 75 core elevations. $F_Q^W(Z)$ evaluations are not applicable for axial core regions, measured in percent of core height:

a. Lower core region, from 0 to 15% inclusive,

b. Upper core region, from 85 to 100% inclusive,

c. Grid plane regions, ± 2% inclusive, and

d. Core plane regions, within ± 2% of the bank demand position of the control banks.

These regions of the core are excluded from the evaluation because of the low probability that they would be more limiting in the safety analyses and because of the difficulty of making a precise measurement in these regions. The excluded regions at the top and bottom of the core are specified in the COLR and are defined to ensure that the minimum margin location is adequately surveilled. A slightly smaller exclusion zone may be specified, if necessary, to include the limiting margin location in the surveilled region of the core.

SR 3.2.1.2 requires a Surveillance of $F_Q^W(Z)$ during the initial startup following each refueling within 24 hours after exceeding 75% RTP. THERMAL POWER levels below 75% are typically non-limiting with respect to the limit for $F_Q^W(Z)$. Furthermore, startup physics testing and flux symmetry measurements, also performed at low power, provide confirmation that the core is operating as expected. This Frequency ensures that verification of $F_Q^W(Z)$ is performed prior to extended operation at power levels where the maximum permitted peak LHR could be challenged and that the first required performance of SR 3.2.1.2 after a refueling is performed at a power level high enough to provide a high level of confidence in the accuracy of the Surveillance result.

Equilibrium conditions are achieved when the core is sufficiently stable at the intended operating conditions required to perform the Surveillance.

If a previous Surveillance of $F_Q^W(Z)$ was performed at part power conditions, SR 3.2.1.2 also requires the $F_Q^W(Z)$ be verified at power levels ≥ 10% RTP above the THERMAL POWER of its last verification within 24 hours after achieving equilibrium conditions. This ensures that $F_Q^W(Z)$ is within its limit using radial peaking factors measured at the higher power level.

BASES

SURVEILLANCE REQUIREMENTS

SR 3.2.1.2 (continued)

The allowance of up to 24 hours after achieving equilibrium conditions will provide a more accurate measurement of $F_Q^W(Z)$ by allowing sufficient time to achieve equilibrium conditions and obtain the power distribution measurement.

The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

REFERENCES

1. 10 CFR 50.46, 1974.
2. USAR, Section 15.4.8.
3. 10 CFR 50, Appendix A, GDC 26.
4. WCAP-7308-L-P-A, "Evaluation of Nuclear Hot Channel Factor Uncertainties," June 1988.
5. Performance Improvement Request 2005-3311.
6. WCAP-12472-P-A, “BEACON Core Monitoring and Operations Support System,” August 1994 (including Addendum 4, September 2012).
7. WCAP-17661-P-A, Revision 1, “Improved RAOC and CAOC FQ Surveillance Technical Specifications,” February 2019.

BASES

APPLICABLE SAFETY ANALYSES (continued)

damage RCS supports, core cooling equipment or core internals. This concern was first identified as Multiplant Action (MPA) D-10 and subsequently as Unresolved Safety Issue (USI) A-2, “Asymmetric LOCA Loads.” This issue was discussed in Reference 4.

The resolution of USI A-2 for Westinghouse PWRs was the use of fracture mechanics technology for RCS piping > 10 inches diameter (Ref. 5). This technology became known as leak-before-break (LBB). Included within the LBB methodology was the requirement to have leak detection systems capable of detecting a 1.0 gpm leak within four hours. This leakage rate is designed to ensure that adequate margins exist to detect leaks in a timely manner during normal operating conditions.

The need to evaluate the severity of an alarm or an indication is important to the operators, and the ability to compare and verify with indications from other systems is necessary. The system response times and sensitivities are described in the USAR (Ref. 3). Multiple instrument locations are utilized, if needed, to ensure that the transport delay time of the leakage from its source to an instrument location yields an acceptable overall response time.

The safety significance of RCS LEAKAGE varies widely depending on its source, rate, and duration. Therefore, detecting and monitoring RCS LEAKAGE into the containment area is necessary. Quickly separating the identified LEAKAGE from the unidentified LEAKAGE provides quantitative information to the operators, allowing them to take corrective action should a leak occur detrimental to the safety of the unit and the public.

RCS leakage detection instrumentation satisfies Criterion 1 of 10 CFR 50.36(c)(2)(ii).

LCO

One method of protecting against large RCS leakage derives from the ability of instruments to rapidly detect extremely small leaks. This LCO requires instruments of diverse monitoring principles to be OPERABLE to provide a high degree of confidence that extremely small leaks are detected in time to allow actions to place the plant in a safe condition, when RCS LEAKAGE indicates possible RCPB degradation.

The LCO is satisfied when monitors of diverse measurement means are available. Thus, the Containment Sump Level and Flow Monitoring System, one containment atmosphere particulate radioactivity monitor and the Containment Cooler Condensate Flow Monitoring System provide an acceptable minimum.

The Containment Sump Level and Flow Monitoring System is considered OPERABLE when it is capable of collecting, measuring and transmitting RCS LEAKAGE to the installed control room indication. The plant

BASES

LCO (continued)

process computer calculates leakage rate from increases in sump level. The leak rate may also be calculated manually once per shift during periods when the plant process computer is out of service.

For containment atmosphere particulate radioactivity monitor instrumentation, OPERABILITY involves more than OPERABILITY of the channel electronics. OPERABILITY also requires correct valve lineups, sample pump operation, sample line insulation and heat tracing, as well as detector OPERABILITY, since these supporting features are necessary for the monitors to rapidly detect RCS LEAKAGE.

The Containment Cooler Condensate Flow Monitoring System is considered OPERABLE when it is capable of measuring liquid flow from the containment coolers. An OPERABLE Containment Cooler Condensate Flow Monitoring System consists of a containment cooler drain collection header, a vertical standpipe, valving, and standpipe level indication for each cooler. Additionally, the plant process computer must be capable of calculating the leakage rate indicated by the condensate being collected. At least two containment coolers must be operating for the Containment Cooler Condensate Flow Monitoring System to be OPERABLE (Ref. 6).

Each containment cooler's air inlet temperature, relative humidity and dewpoint determines the amount of condensate the cooler will produce. Historically, the "D" containment cooler has produced the least amount of condensate. A containment cooler that is producing a small amount of condensate may result in a stable standpipe level and infrequent actuation of the automatic dump valve. Indications that the automatic dump valve may not be operating properly are no level indication in the standpipe or overfilling of the containment cooling drip pan (Ref. 7).

APPLICABILITY

Because of elevated RCS temperature and pressure in MODES 1, 2, 3, and 4, RCS leakage detection instrumentation is required to be OPERABLE.

In MODE 5 or 6, the temperature is required to be ≤ 200°F and pressure is maintained low or at atmospheric pressure. Since the temperatures and pressures are far lower than those for MODES 1, 2, 3, and 4, the likelihood of leakage and crack propagation are much smaller. Therefore, the requirements of this LCO are not applicable in MODES 5 and 6.

ACTIONS

A.1 and A.2

A primary system leak would result in reactor coolant flowing into the containment normal sumps or into the instrument tunnel sump. Indication of increasing sump level is transmitted to the control room by means of

BASES

ACTIONS

A.1 and A.2 (continued)

individual sump level transmitters. This information is used to provide measurement of low leakage by monitoring level increase versus time. With the required Containment Sump Level and Flow Monitoring System inoperable, no other form of sampling can provide the equivalent information; however, the containment atmosphere particulate radioactivity monitor will provide indications of changes in leakage. Together with the atmosphere monitor, the periodic surveillance for RCS water inventory balance, SR 3.4.13.1, must be performed at an increased frequency of 24 hours to provide information that is adequate to detect leakage. A Note is added allowing that SR 3.4.13.1 is not required to be performed until 12 hours after establishing steady state operation (stable RCS pressure, temperature, power level, pressurizer and makeup tank level, makeup and letdown, and RCP seal injection and return flows). The 12 hour allowance provides sufficient time to collect and process all necessary data after stable plant conditions are established.

Restoration of the required Containment Sump Level and Flow Monitoring System to OPERABLE status within a Completion Time of 30 days is required to regain the function after the system's failure. This time is acceptable, considering the Frequency and adequacy of the RCS water inventory balance required by Required Action A.1. The Completion Time is modified by a Note indicating that the 30 days is extended until startup from a plant shutdown or startup from Refueling Outage 20.

B.1.1, B.1.2, B.2.1 and B.2.2

With the containment atmosphere particulate radioactivity monitoring instrumentation channel inoperable, alternative action is required. Either samples of the containment atmosphere must be taken and analyzed for particulate radioactivity or water inventory balances, in accordance with SR 3.4.13.1, must be performed to provide alternate periodic information. Alternatively, continued operation is allowed if the containment air cooler condensate monitoring system is OPERABLE, provided grab samples are taken or water inventory balances are performed every 24 hours.
With a sample obtained and analyzed or water inventory balance performed every 24 hours, the reactor may be operated for up to 30 days to allow restoration of the required containment atmosphere particulate radioactivity monitor.

The 24 hour interval provides periodic information that is adequate to detect leakage. A Note is added allowing that SR 3.4.13.1 is not required to be performed until 12 hours after establishing steady state operation (stable RCS pressure, temperature, power level, pressurizer and makeup tank level, makeup and letdown, and RCP seal injection and return flows). The 12 hour allowance provides sufficient time to collect and process all necessary data after stable plant conditions are established. The 30 day

BASES

ACTIONS

B.1.1, B.1.2, B.2.1 and B.2.2 (continued)

Completion Time recognizes at least one other form of leakage detection is available.

C.1 and C.2

With the required containment cooler condensate monitoring system inoperable, alternative action is again required. Either SR 3.4.15.1 must be performed or water inventory balances, in accordance with SR 3.4.13.1, must be performed to provide alternate periodic information. Provided a CHANNEL CHECK is performed every 8 hours or a water inventory balance is performed every 24 hours, reactor operation may continue while awaiting restoration of the containment cooler condensate monitoring system to OPERABLE status.

The 24 hour interval provides periodic information that is adequate to detect RCS LEAKAGE. A Note is added allowing that SR 3.4.13.1 is not required to be performed until 12 hours after establishing steady state operation (stable RCS pressure, temperature, power level, pressurizer and makeup tank level, makeup and letdown, and RCP seal injection and return flows.) The 12 hour allowance provides sufficient time to collect and process all necessary data after stable plant conditions are established.

D.1 and D.2

With the required containment atmosphere particulate radioactivity monitor and the required Containment Cooler Condensate Monitoring System inoperable, the means of detecting leakage is the Containment Sump Level and Flow Monitoring System. This Condition does not provide all the required diverse means of leakage detection. The Required Action is to restore either of the inoperable required monitoring methods to OPERABLE status within 30 days to regain the intended leakage detection diversity. The 30 day Completion Time ensures that the plant will not be operated in a reduced configuration for a lengthy time period.

Refer to LCO 3.3.6, “Containment Purge Isolation Instrumentation,” upon a loss of the required containment atmosphere radioactivity monitor to ensure LCO requirements are met.

BASES

ACTIONS (continued)

E.1 and E.2

If a Required Action of Condition A, B, C or D cannot be met, the plant must be brought to a MODE in which the requirement does not apply. To achieve this status, the plant must be brought to at least MODE 3 within 6 hours and to MODE 5 within 36 hours. The allowed Completion Times are reasonable, based on operating experience, to reach the required plant conditions from full power conditions in an orderly manner and without challenging plant systems.

F.1

With all required monitoring methods inoperable, no automatic means of monitoring leakage are available, and immediate plant shutdown in accordance with LCO 3.0.3 is required.

SURVEILLANCE REQUIREMENTS

SR 3.4.15.1

SR 3.4.15.1 requires the performance of a CHANNEL CHECK of the required containment atmosphere particulate radioactivity monitor. The check gives reasonable confidence that the channel is operating properly. The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

SR 3.4.15.2

SR 3.4.15.2 requires the performance of a COT on the required containment atmosphere particulate radioactivity monitor. The test ensures that the monitor can perform its function in the desired manner. The test verifies the alarm setpoint and relative accuracy of the instrument string. The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

SR 3.4.15.3, SR 3.4.15.4, and SR 3.4.15.5

These SRs require the performance of a CHANNEL CALIBRATION for each of the RCS leakage detection instrumentation channels. The calibration verifies the accuracy of the instrument string, including the instruments located inside containment. The Surveillance Frequency is controlled under the Surveillance Frequency Control Program.

BASES

REFERENCES

1. 10 CFR 50, Appendix A, Section IV, GDC 30.

2. Regulatory Guide 1.45.

3. USAR, Section 5.2.5.

4. NUREG-609, “Asymmetric Blowdown Loads on PWR Primary Systems,” 1981.

5. Generic Letter 84-04, “Safety Evaluation of Westinghouse Topical Reports Dealing with Elimination of Postulated Pipe Breaks in PWR Primary Main Loops.”

6. USAR, Section 6.2.2.2.2.

7. Performance Improvement Request 2005-2823.

8. NRC letter, “Wolf Creek Generating Station – License Amendment Request to Change the Reactor Coolant System Leakage Detection Instrumentation Methodology (TAC No. MC8214),” May 16, 2006.

B 3.8 ELECTRICAL POWER SYSTEMS

B 3.8.1 AC Sources - Operating

BASES

BACKGROUND

The unit Class 1E AC Electrical Power Distribution System AC sources consist of the offsite power sources (preferred power sources, normal and alternate from the redundant Engineered Safety Feature (ESF) transformers), and the onsite standby power sources (Train A and Train B diesel generators (DGs)). As required by 10 CFR 50, Appendix A, GDC 17 (Ref. 1), the design of the AC electrical power system provides independence and redundancy to ensure an available source of power to the ESF systems.

The onsite Class 1E AC Distribution System is divided into redundant load groups (trains) so that the loss of any one group does not prevent the minimum safety functions from being performed. Each train has connections to its preferred offsite power source and a single DG.

Offsite power is supplied to the unit switchyard from the transmission network by four transmission lines. From the switchyard, two electrically and physically separated circuits provide AC power, through the ESF transformers, to the 4.16 kV ESF buses. A detailed description of the offsite power network and the circuits to the Class 1E ESF buses is found in the USAR, Chapter 8 (Ref. 2).

An offsite circuit consists of all breakers, transformers, voltage regulating tap changers (when installed), switches, interrupting devices, cabling, and controls required to transmit power from the offsite transmission network to the onsite Class 1E ESF bus(es).

Certain required unit loads are returned to service in a predetermined sequence in order to prevent overloading the transformer supplying offsite power to the onsite Class 1E Distribution System. Within 1 minute after the initiating signal is received, all automatic and permanently connected loads needed to recover the unit or maintain it in a safe condition are returned to service via the load sequencer.

The onsite standby power source for each 4.16 kV ESF bus is a dedicated DG. DGs A and B are dedicated to ESF buses NB01 and NB02, respectively. The DG starts automatically on a safety injection (SI) signal or on an ESF bus undervoltage signal. A degraded voltage signal produces an undervoltage condition by opening the normal and alternate feeder breakers to the bus(es) experiencing degraded voltage. Both signals are initiated from the load shedder and emergency load sequencer (LSELS). OPERABILITY of the undervoltage and degraded voltage instrumentation functions are addressed in LCO 3.3.5, “Loss of